

Applied Continuous Multivariate Analysis (36-722)

Multivariate Outliers; Principal Components Biplot; Robust PCA

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April 16, 2012

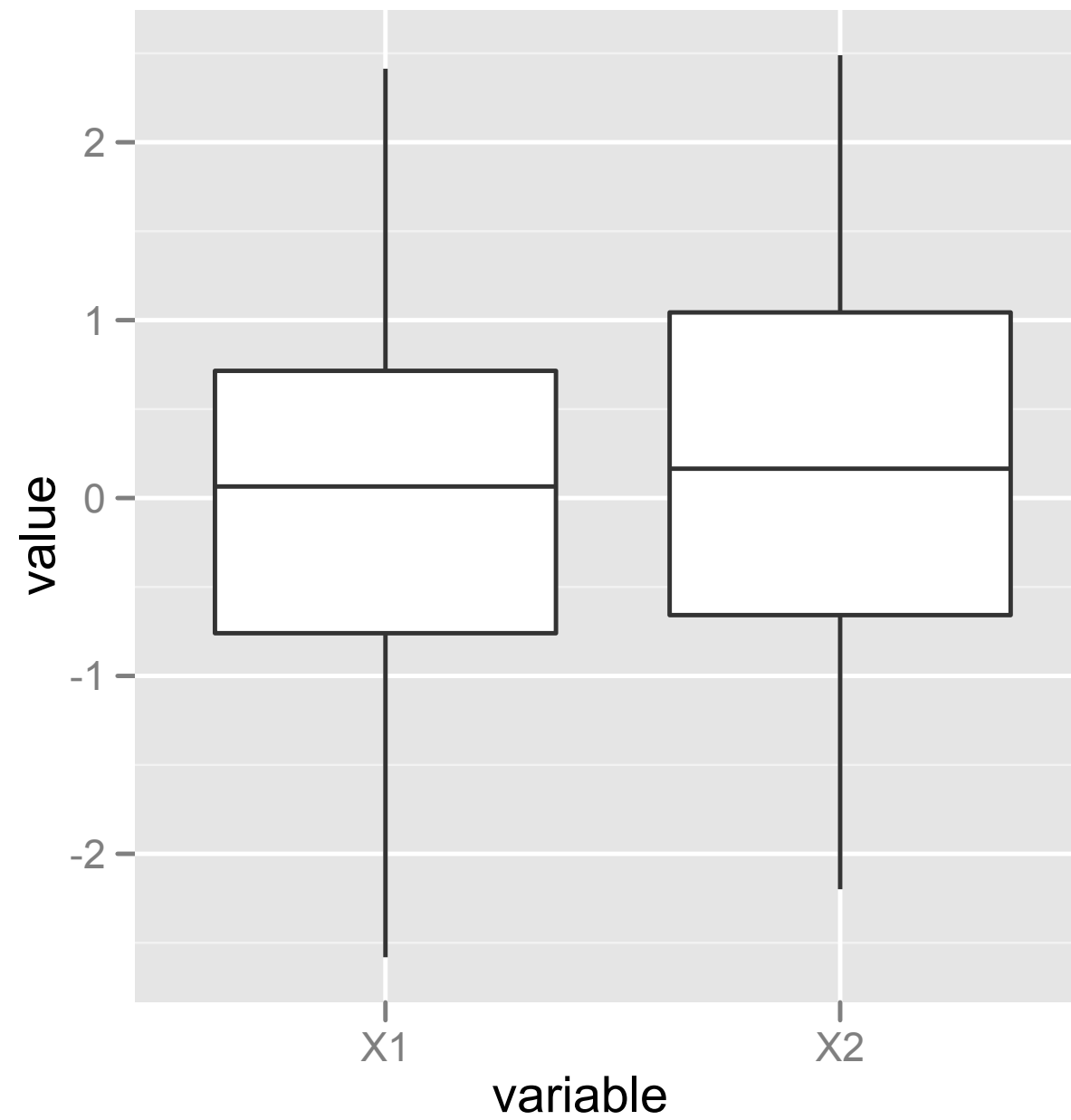
Agenda

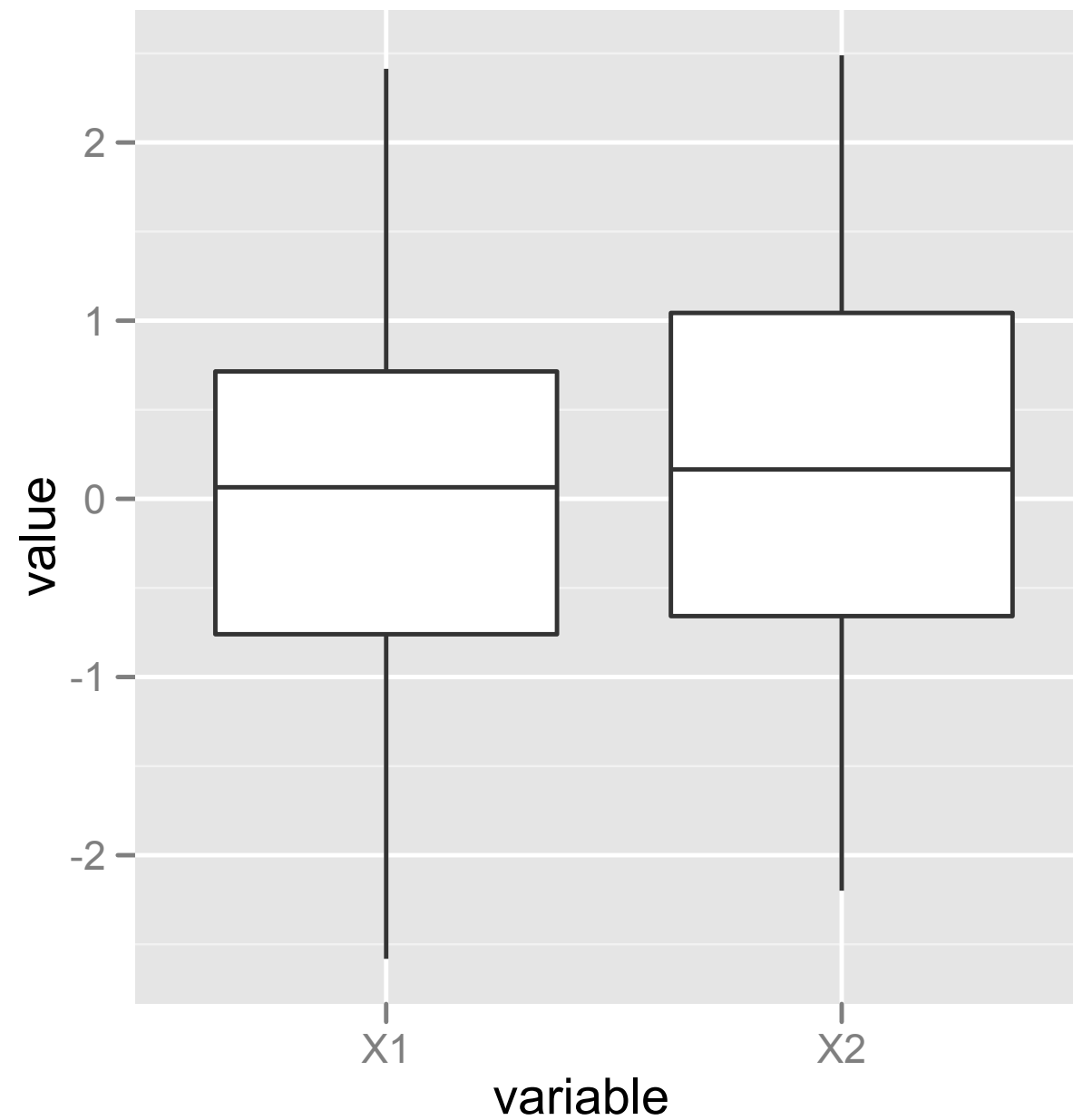
- Multivariate outliers
 - What is a multivariate outlier?
 - Detection
- Principal components biplot
- Robust PCA (time allowing)

Multivariate Outliers

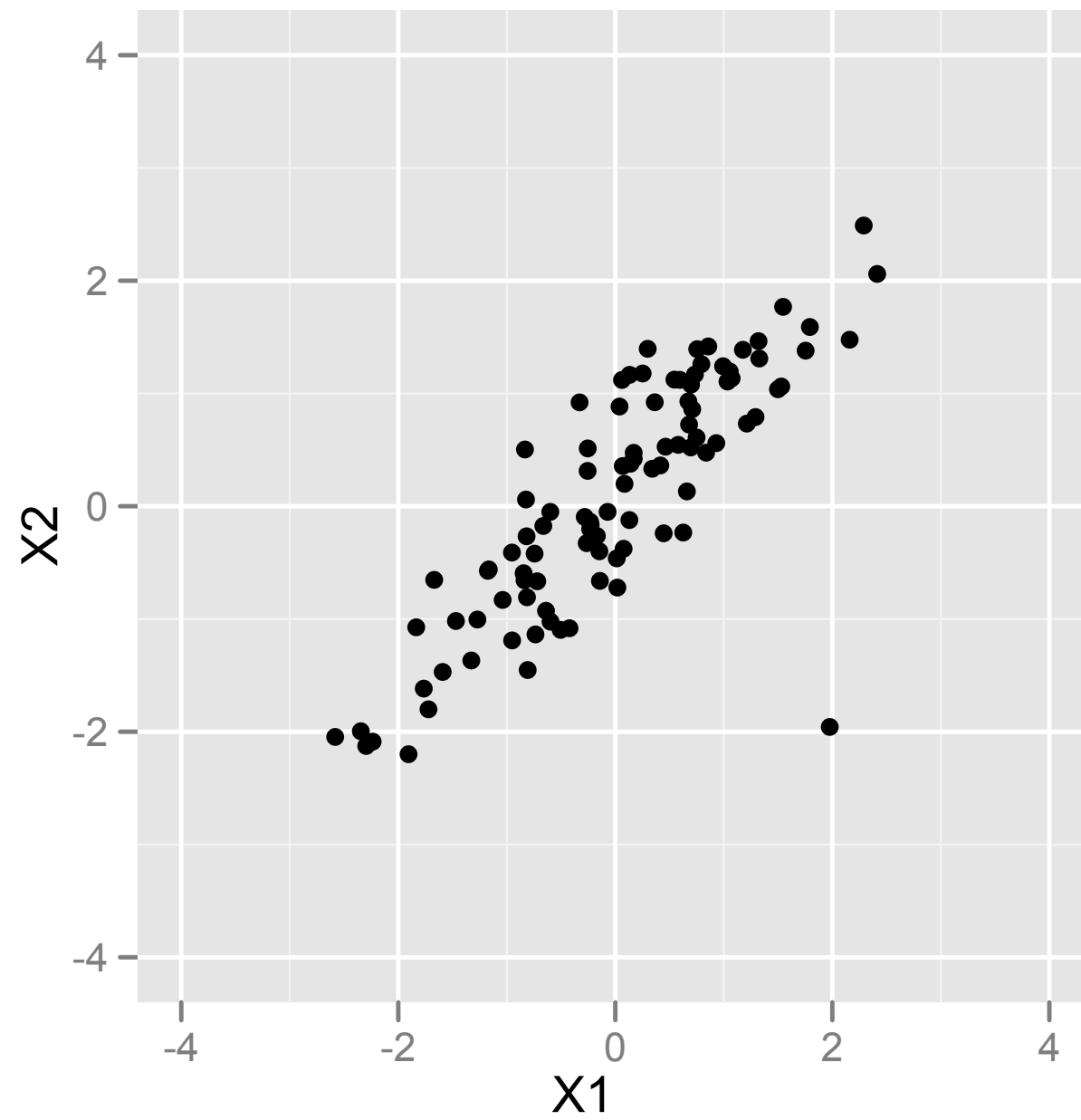
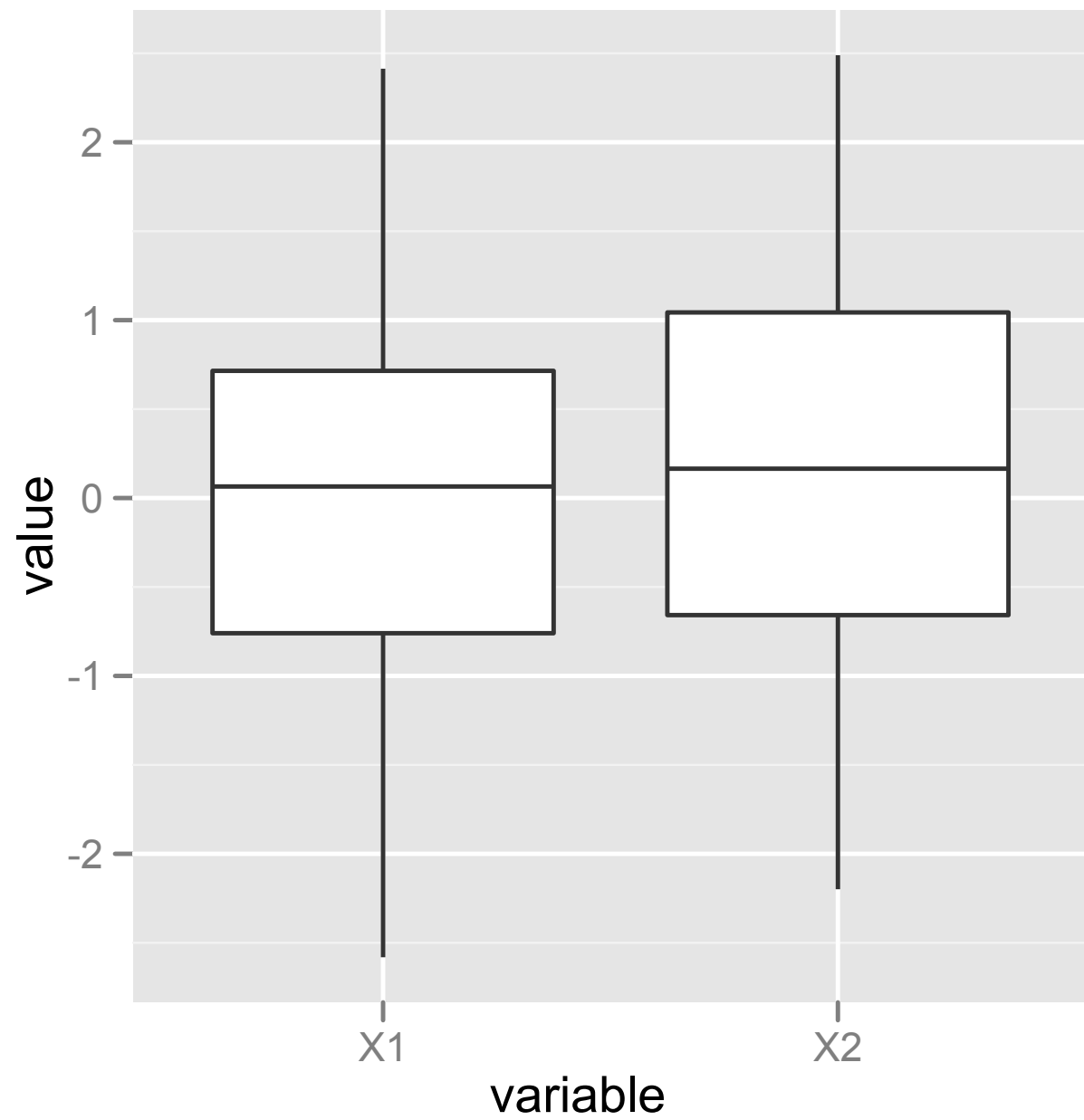
Multivariate Outliers

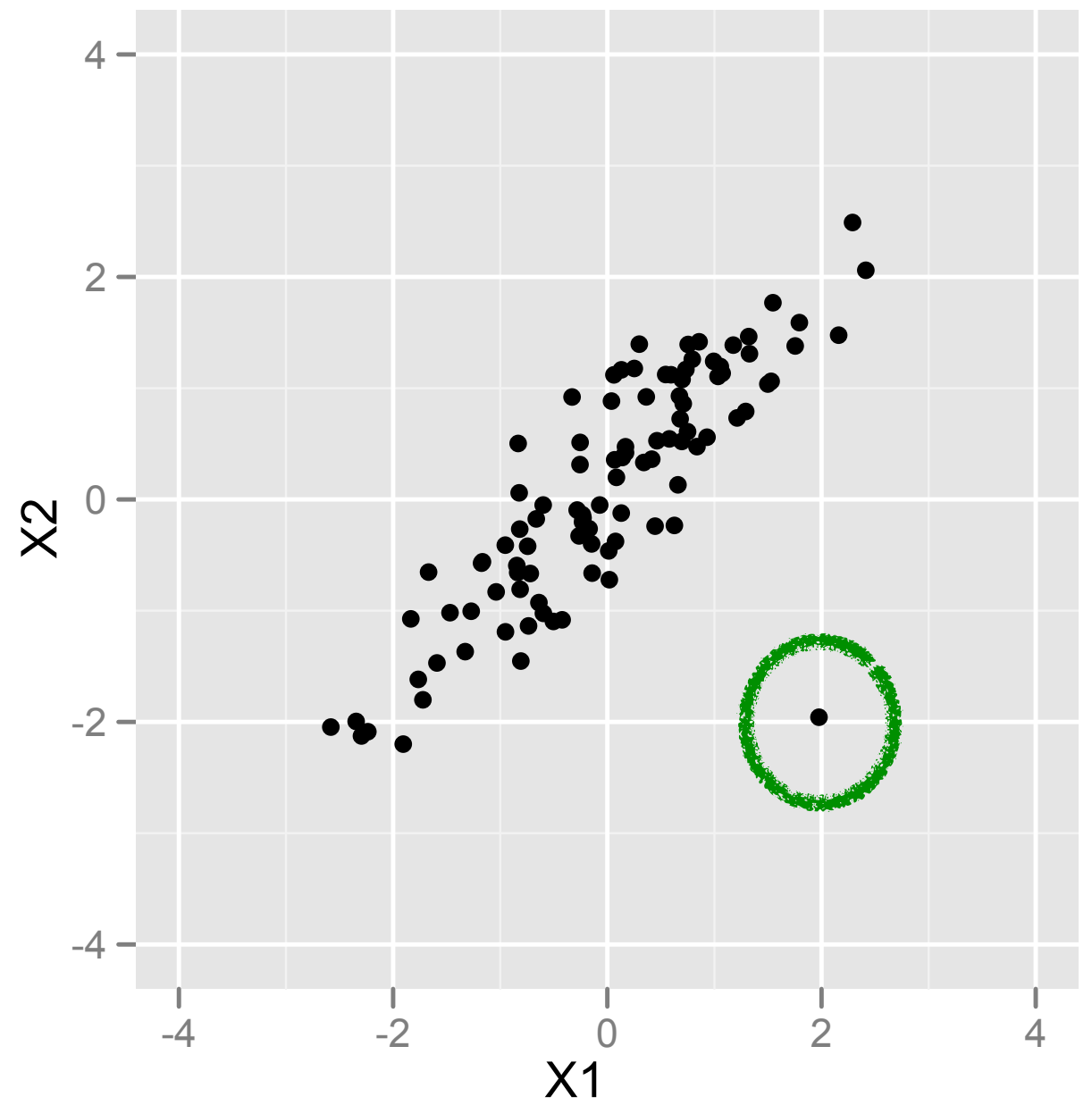
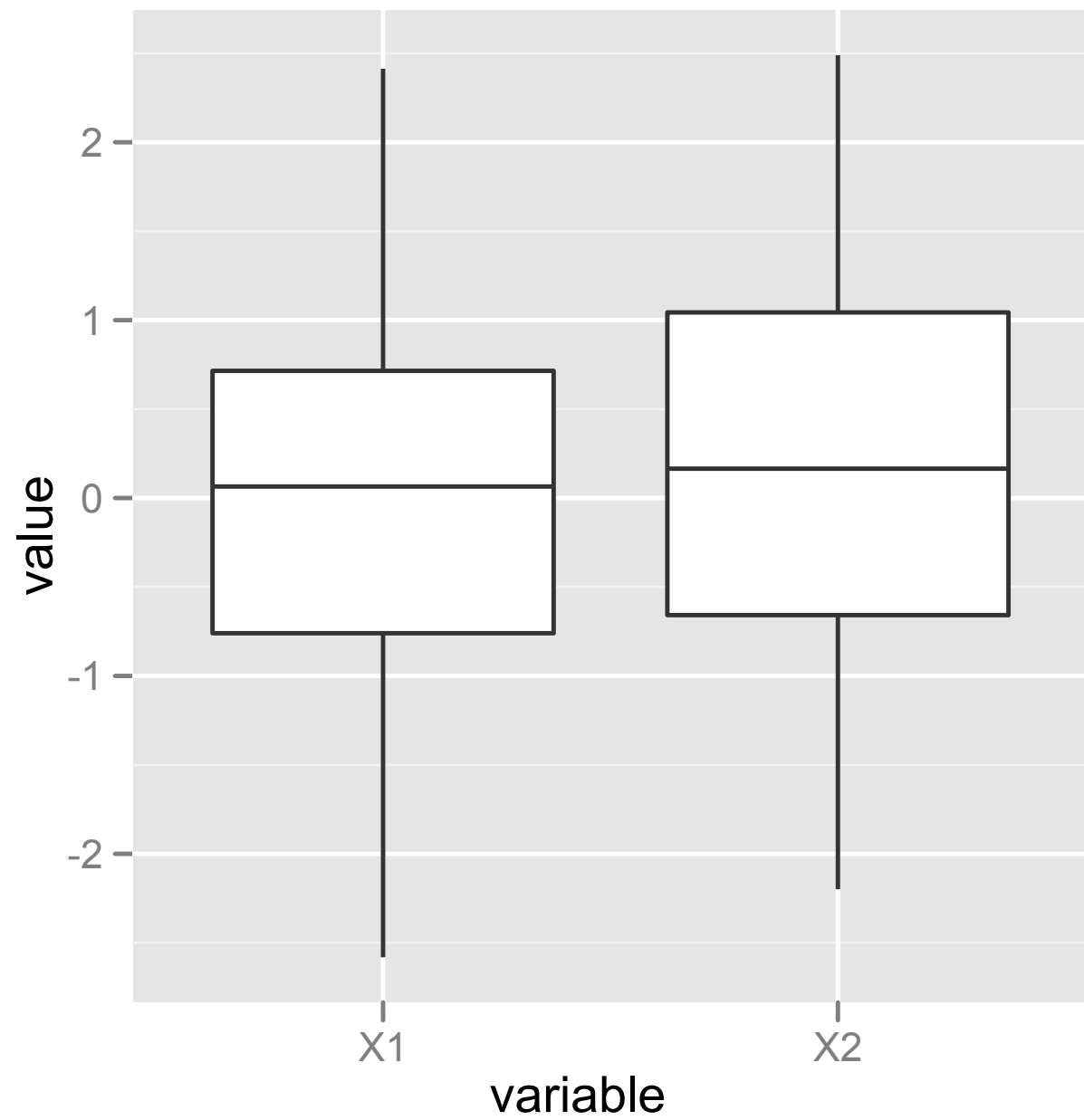
- An outlier is an observation that is ***unusually*** different from most other observations.
- Some outliers cannot be detected by looking individual variables.
- A multivariate outlier is not necessarily a univariate outlier.





No apparent outliers





Outlier?

Multivariate Outlier Detection

- Multivariate outliers often ***disrespect*** the covariance structure of the data.

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Multivariate Outlier Detection

- Multivariate outliers often ***disrespect*** the covariance structure of the data.
- “tall fathers tend to have tall sons”
- “short father, but tall son”
- There are short fathers and tall fathers, but seldom “short father, but tall son”

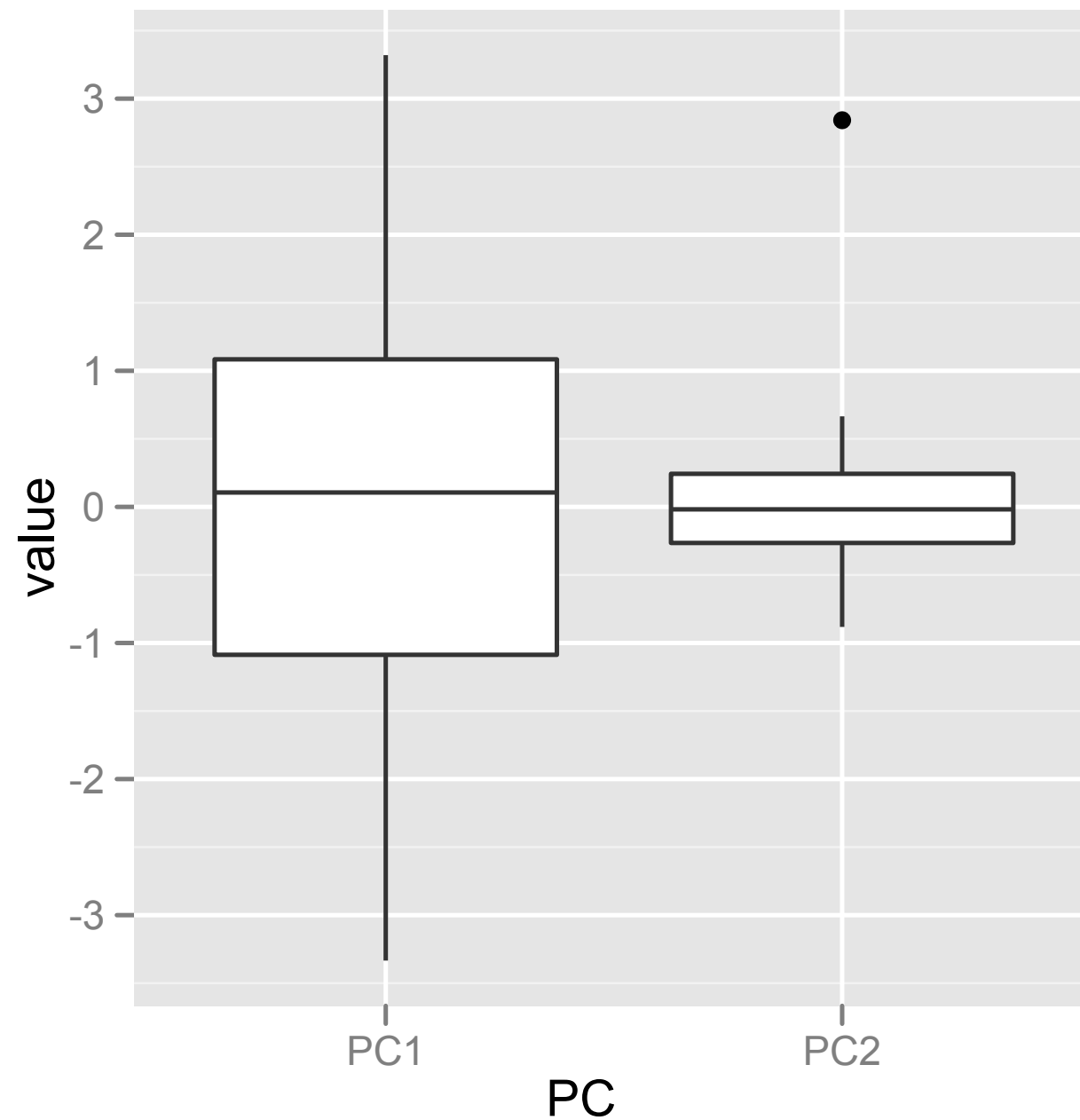
Outlier detection with PCA

- Outliers that are detectable by plots of the first few PCs inflate variances and covariances
- Last few PCs may reveal outliers that disrespect the covariance structure
- Moral: Look at the first and **last** few PCs!

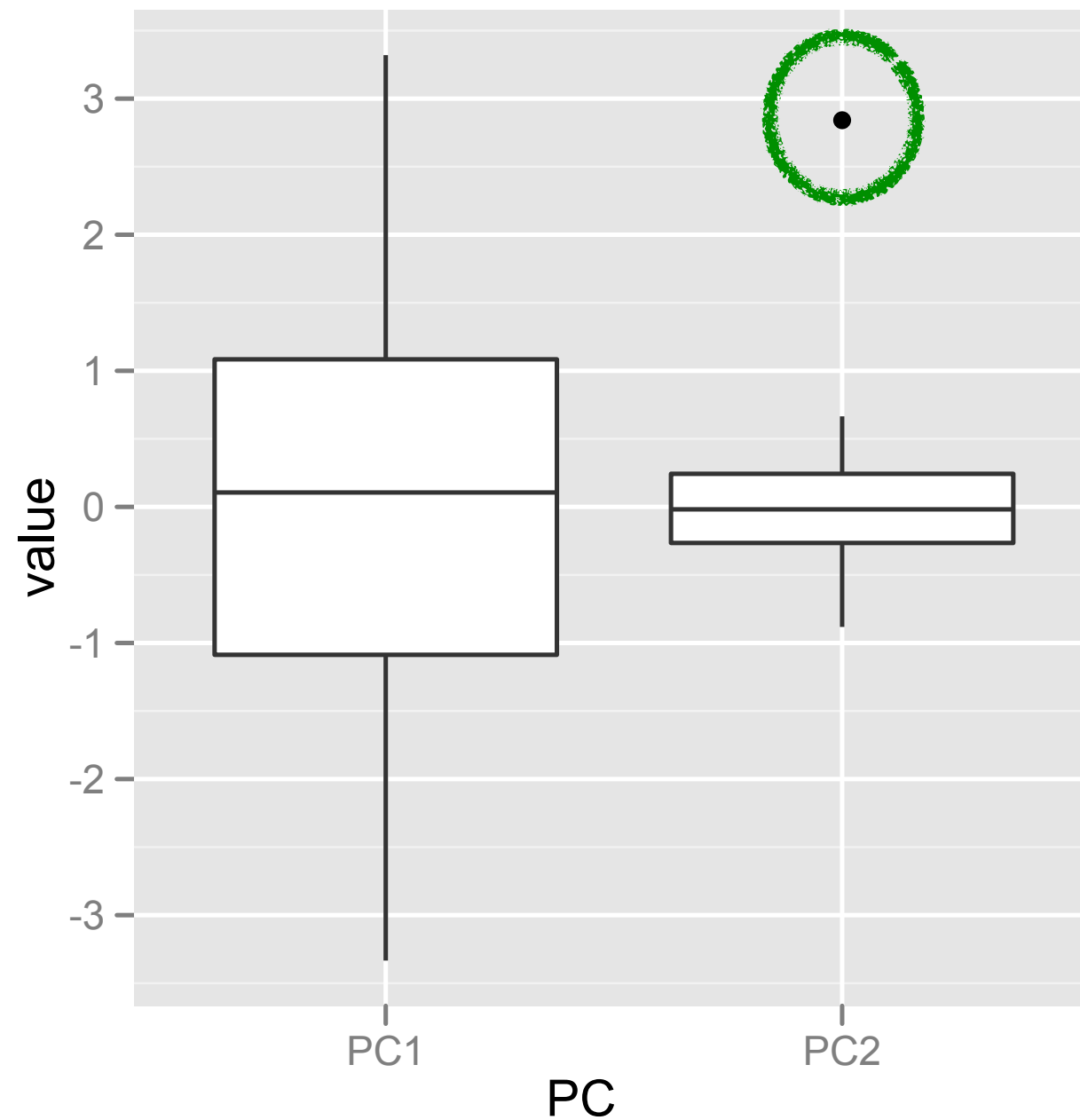
Idea

- Look for outliers among the principal component scores
 - Scatterplots
 - Boxplots
 - Normal Q-Q plots

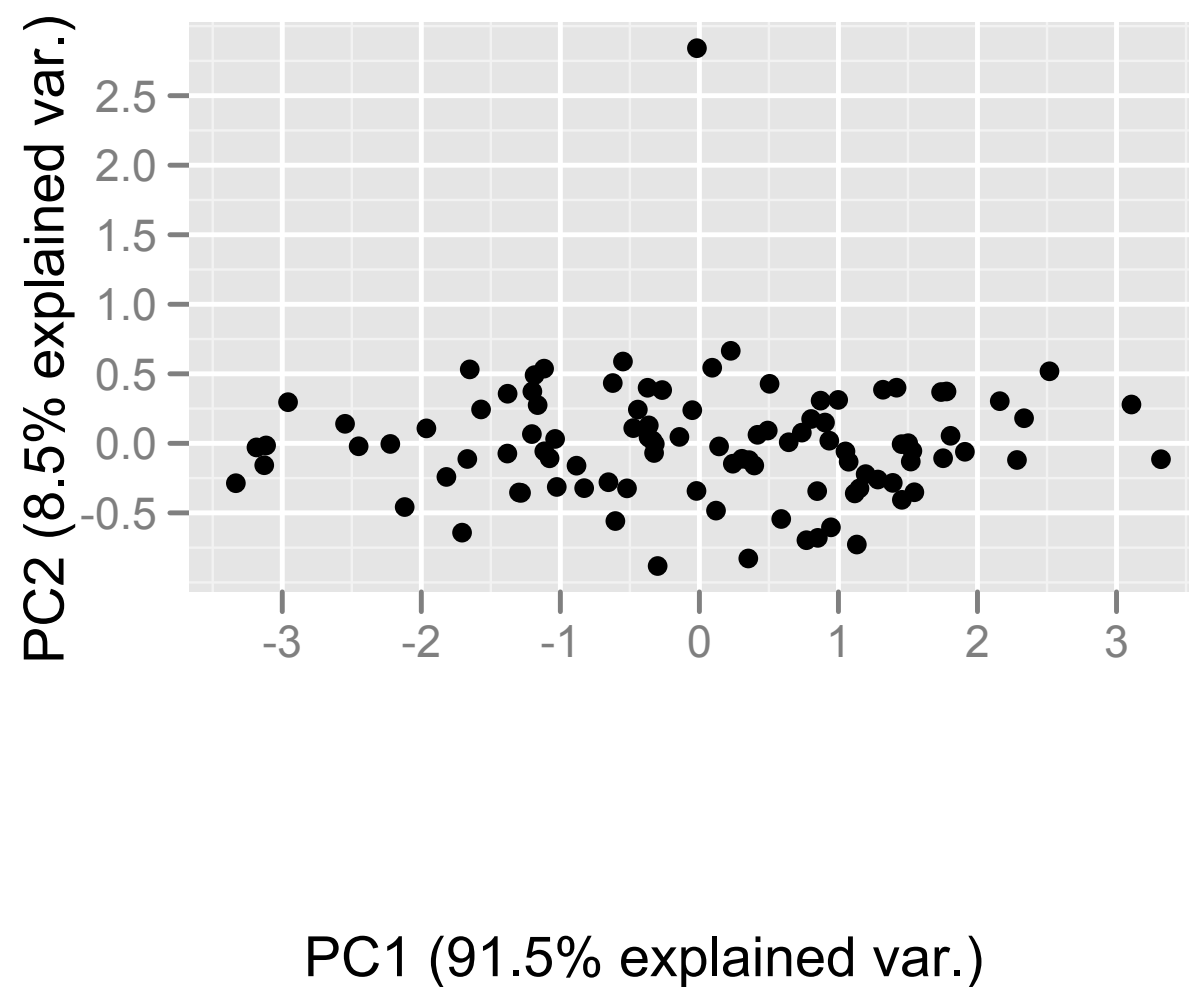
Outlier detection with PCA



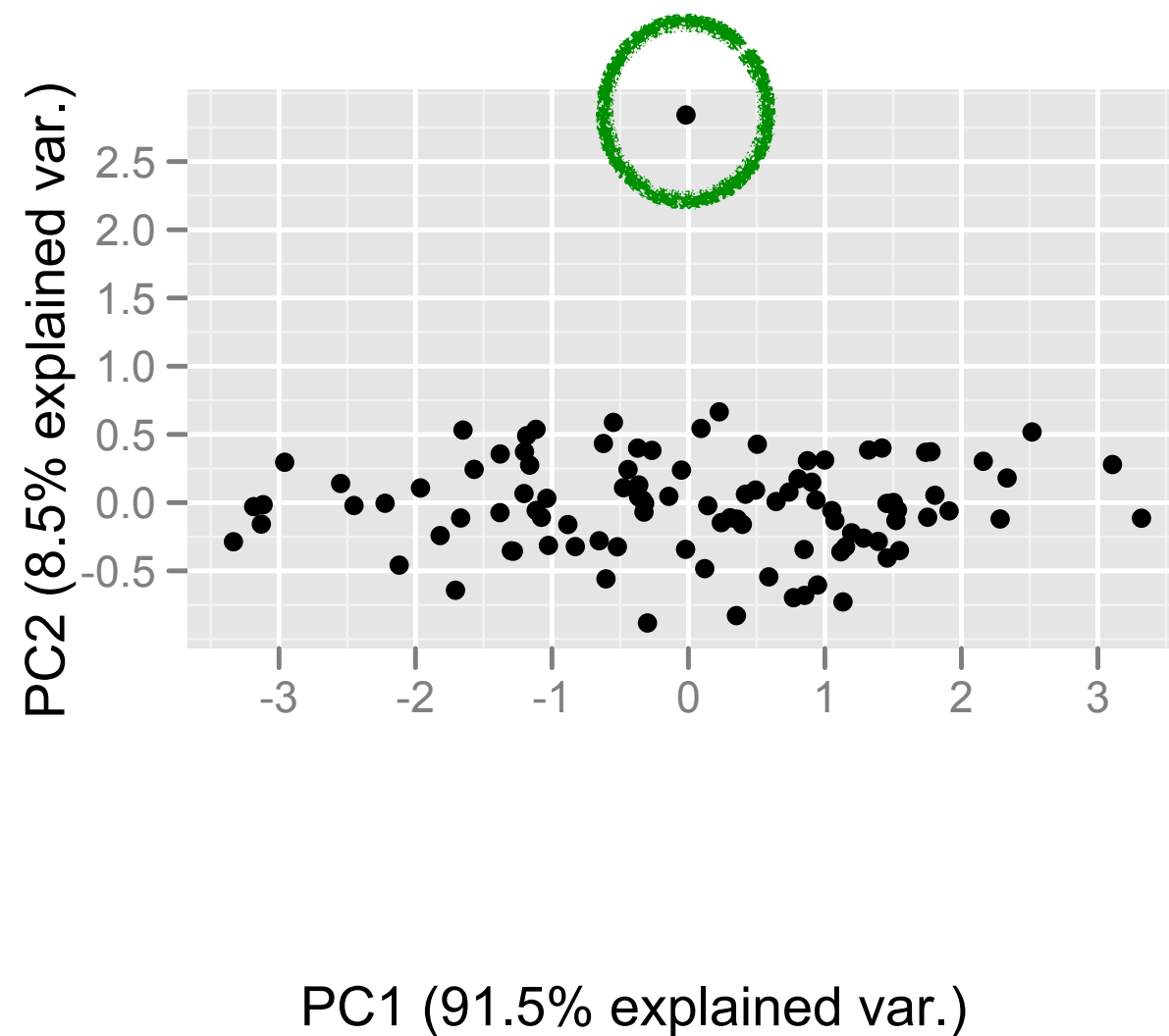
Outlier detection with PCA



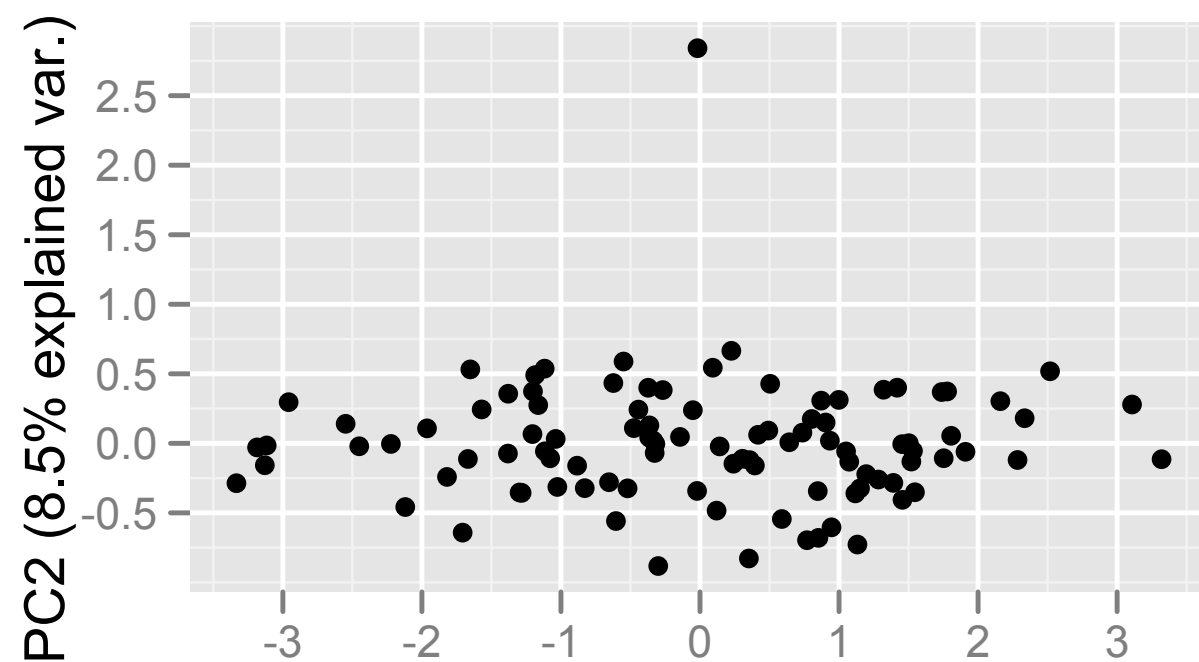
Outlier detection with PCA



Outlier detection with PCA



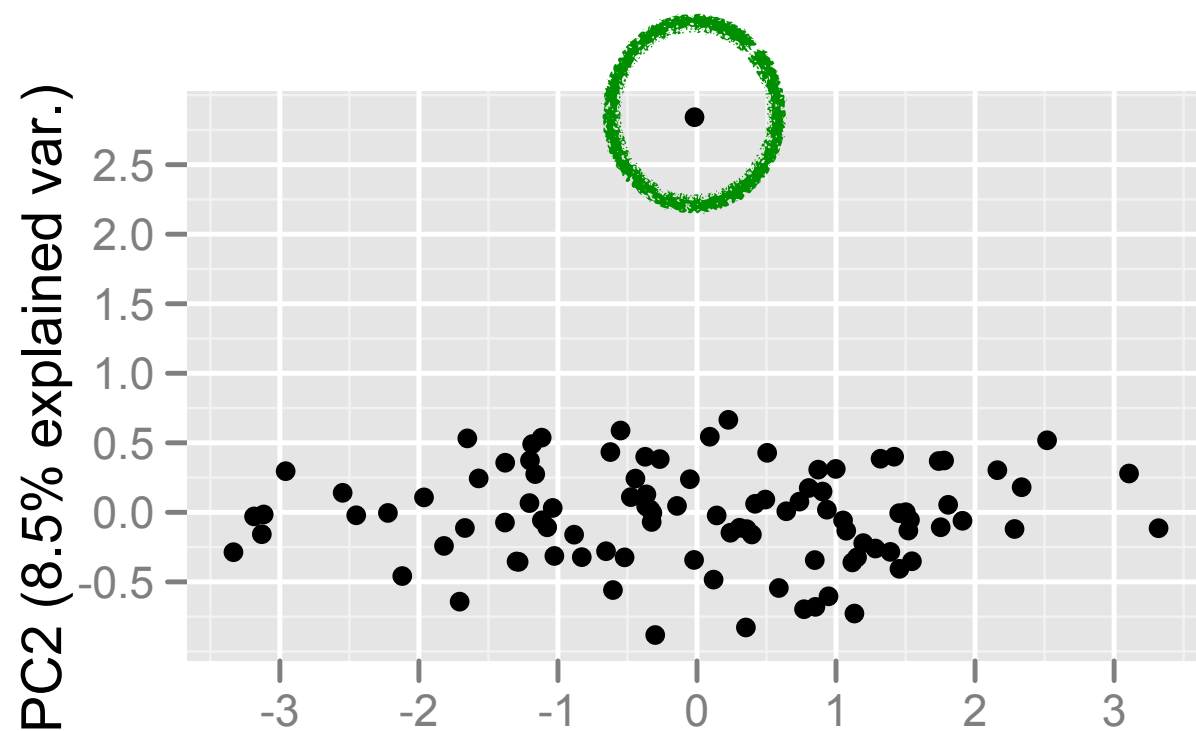
Outlier detection with PCA



```
> which(x.pca$x[,2] > 2.5)
[1] 100
```

PC1 (91.5% explained var.)

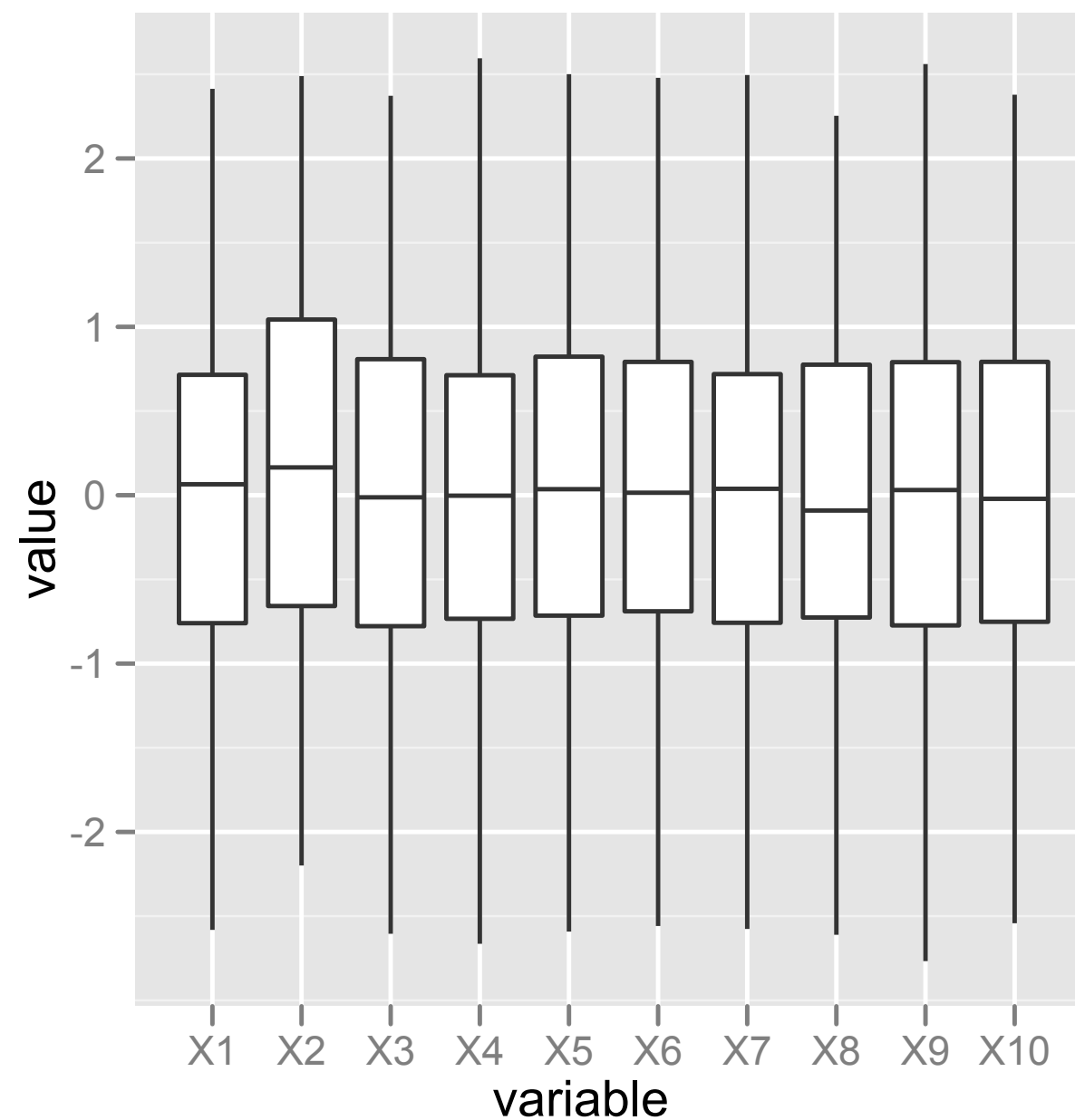
Outlier detection with PCA



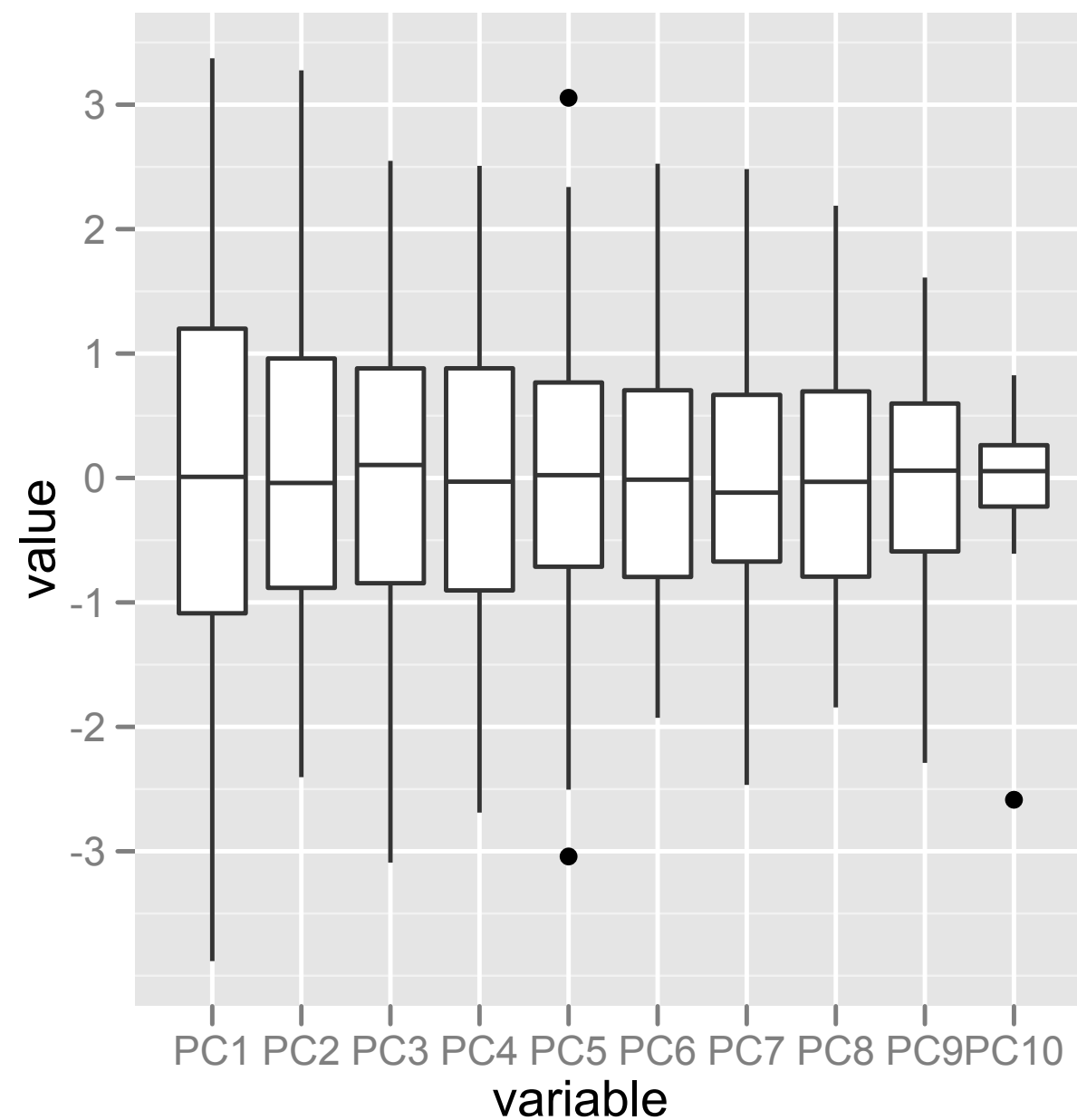
```
> which(x.pca$x[,2] > 2.5)
[1] 100
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PC1 (91.5% explained var.)

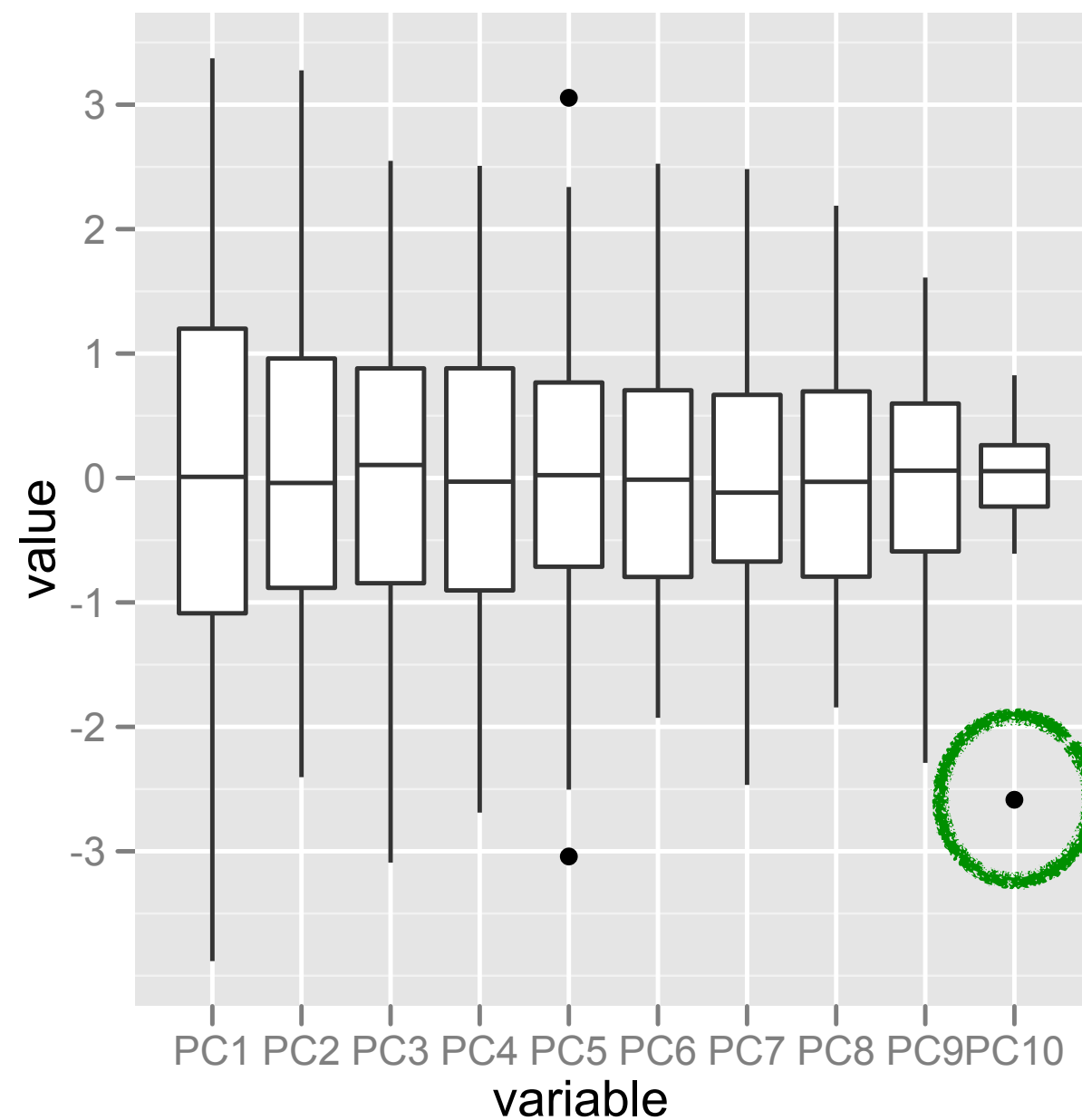
Outlier detection with PCA



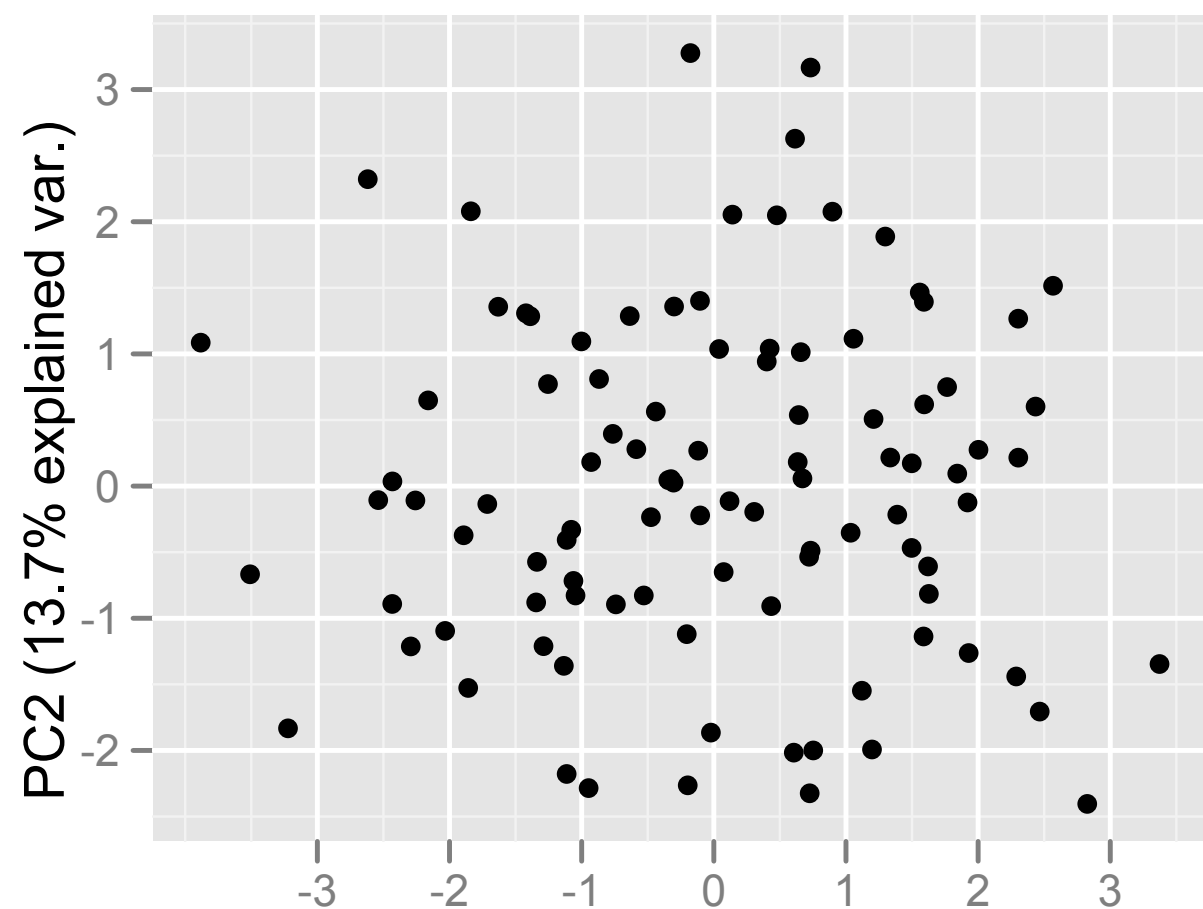
Outlier detection with PCA



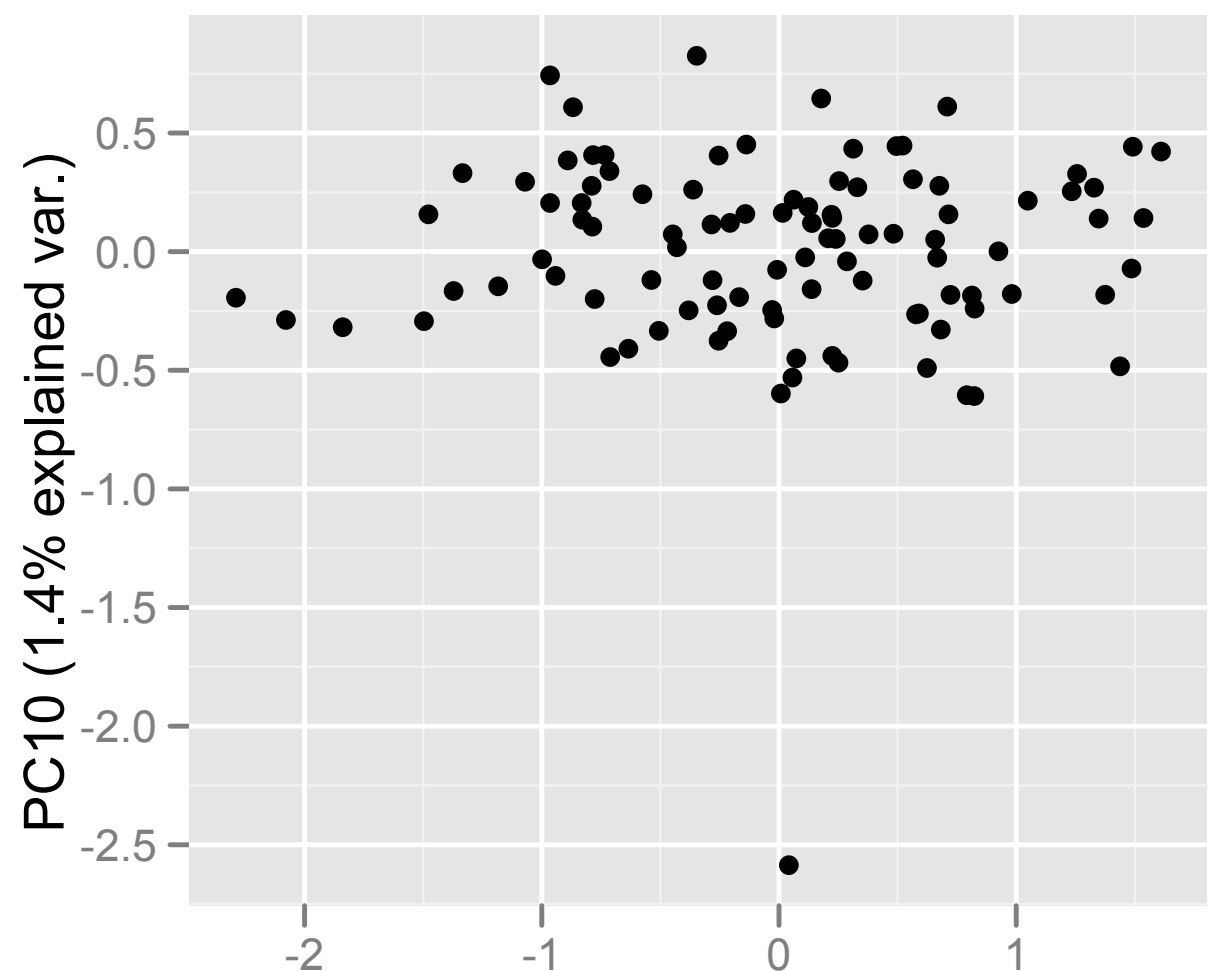
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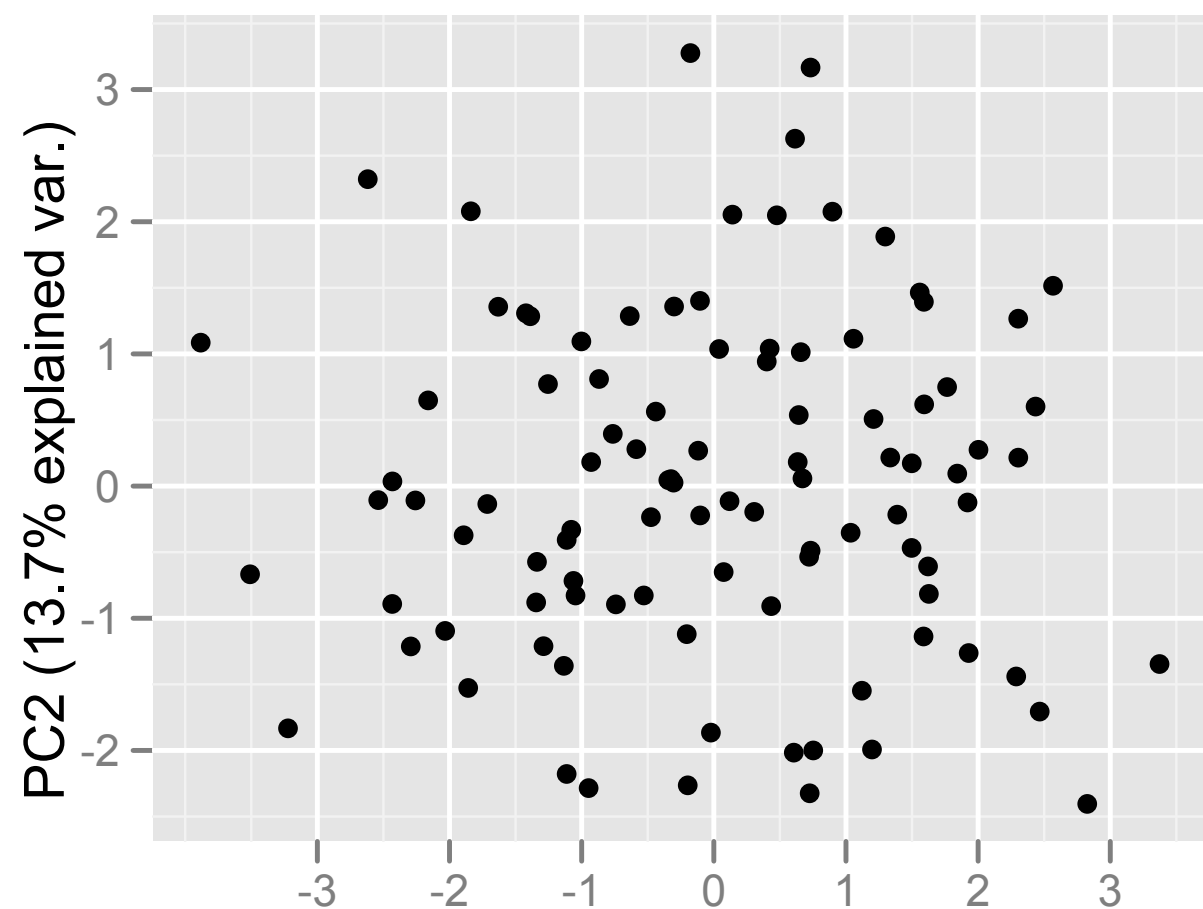


PC1 (19.9% explained var.)

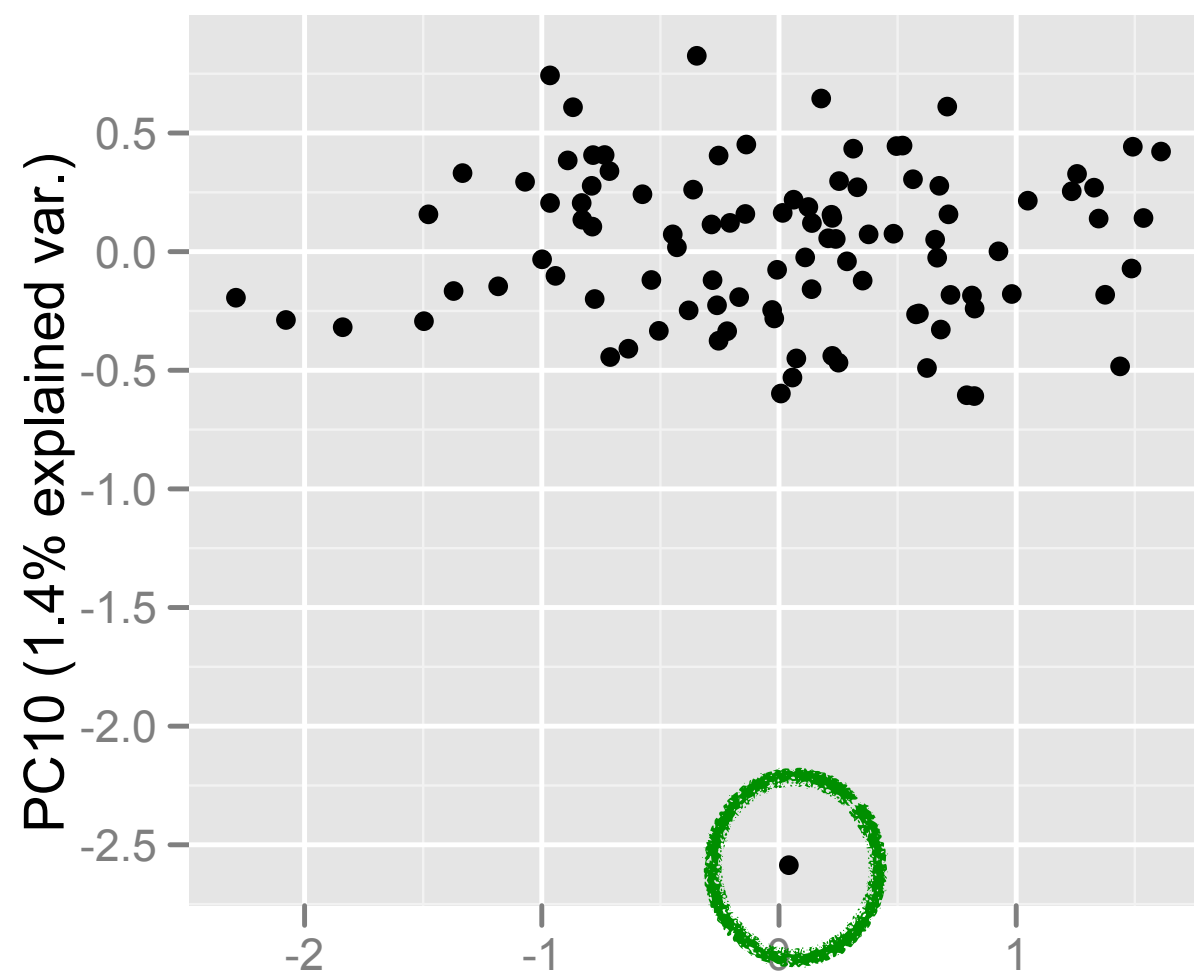


PC9 (5.9% explained var.)

Outlier detection with PCA

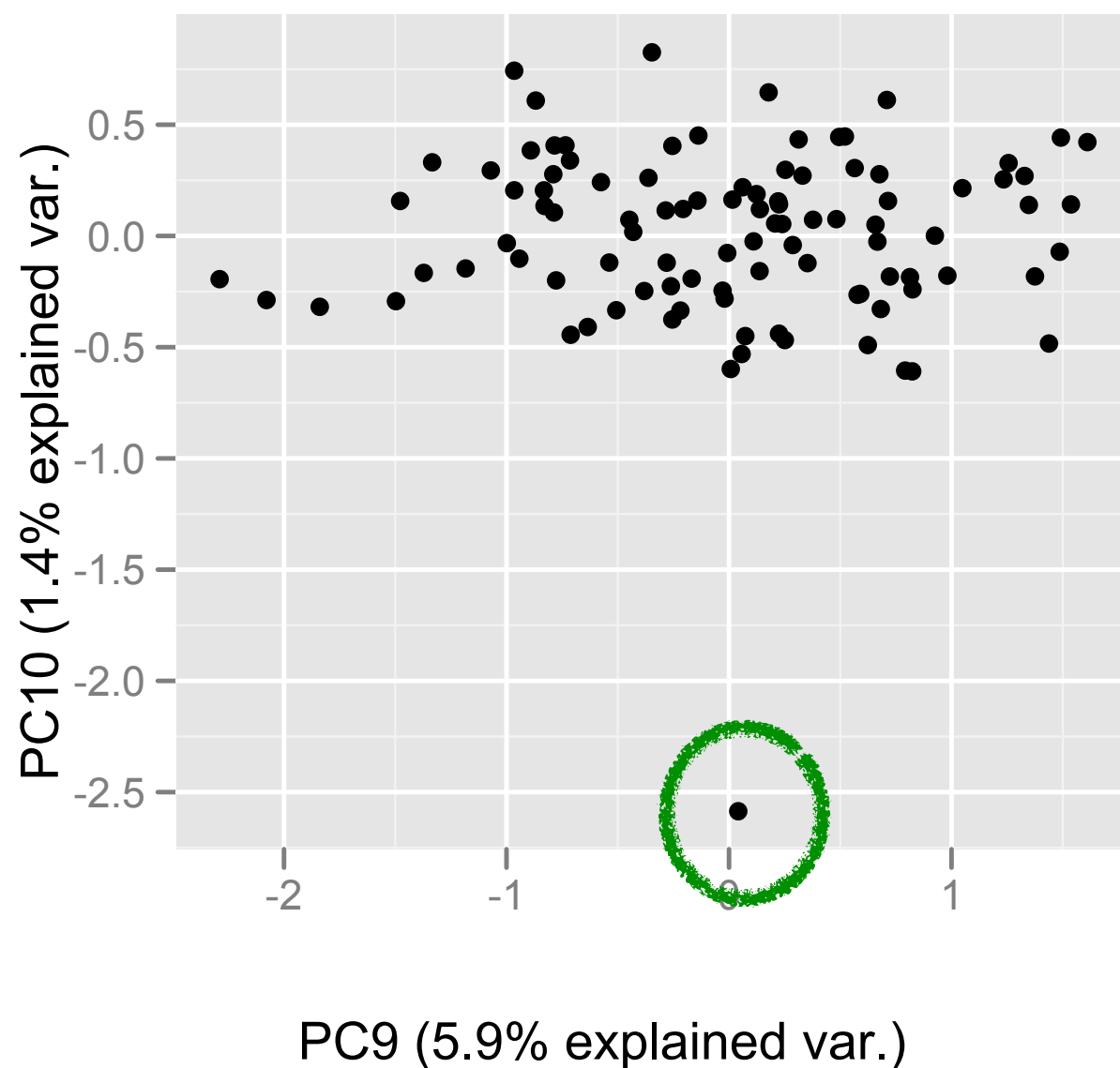


PC1 (19.9% explained var.)



PC9 (5.9% explained var.)

Outlier detection with PCA

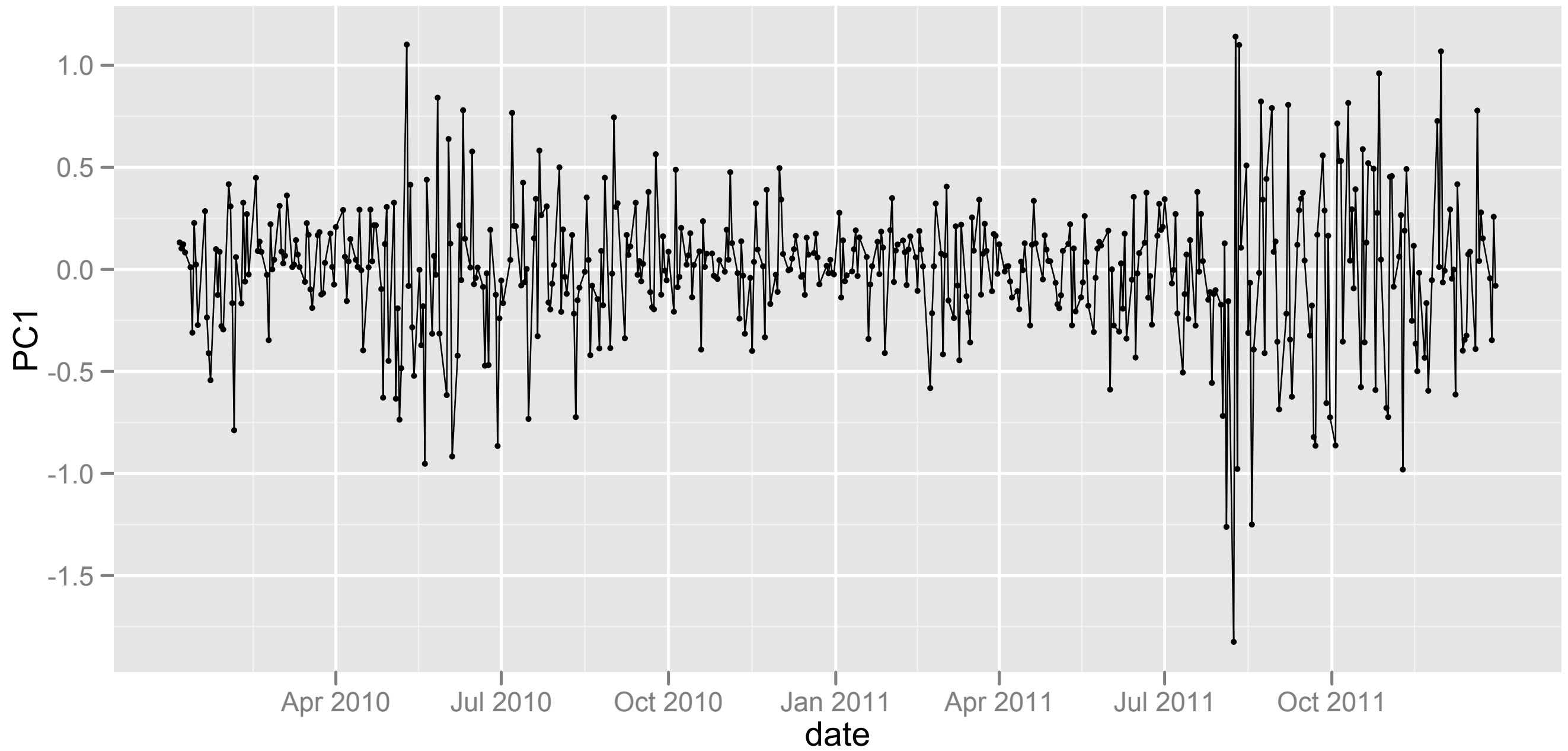


```
> which(x.pca$x[,10] < -2)
[1] 100
```

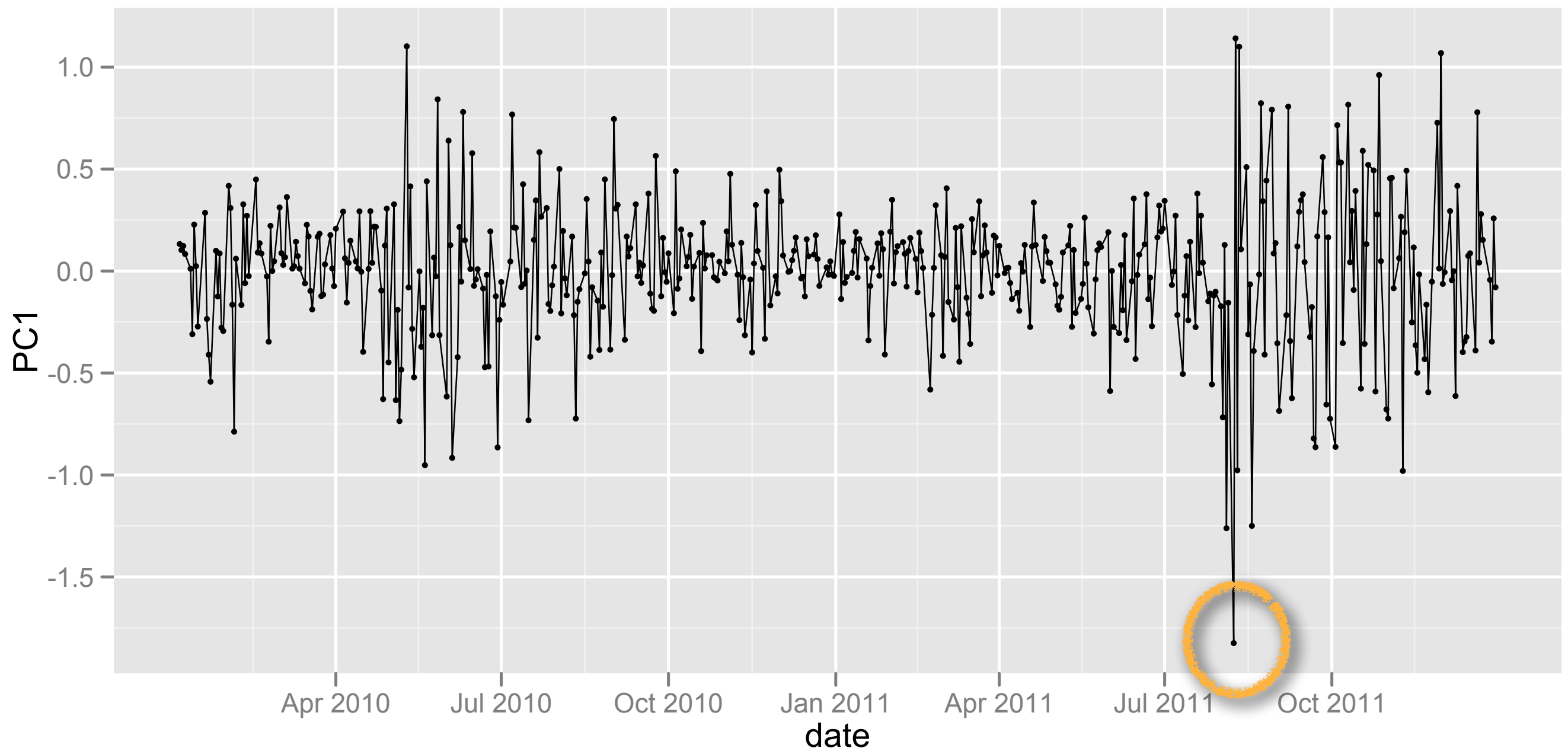
S&P 500 data

- Log daily returns of ≈ 500 constituents of S&P 500 (as of April, 2012) over the period Jan 1, 2010 – Dec 31, 2011
- PCA applied to the sample covariance matrix.
- Recall that PC1 loads positively on all ≈ 500 stocks – reflects positive correlation of between all ≈ 500 stocks.

Principal Component I Scores



Principal Component I Scores



August 8, 2011

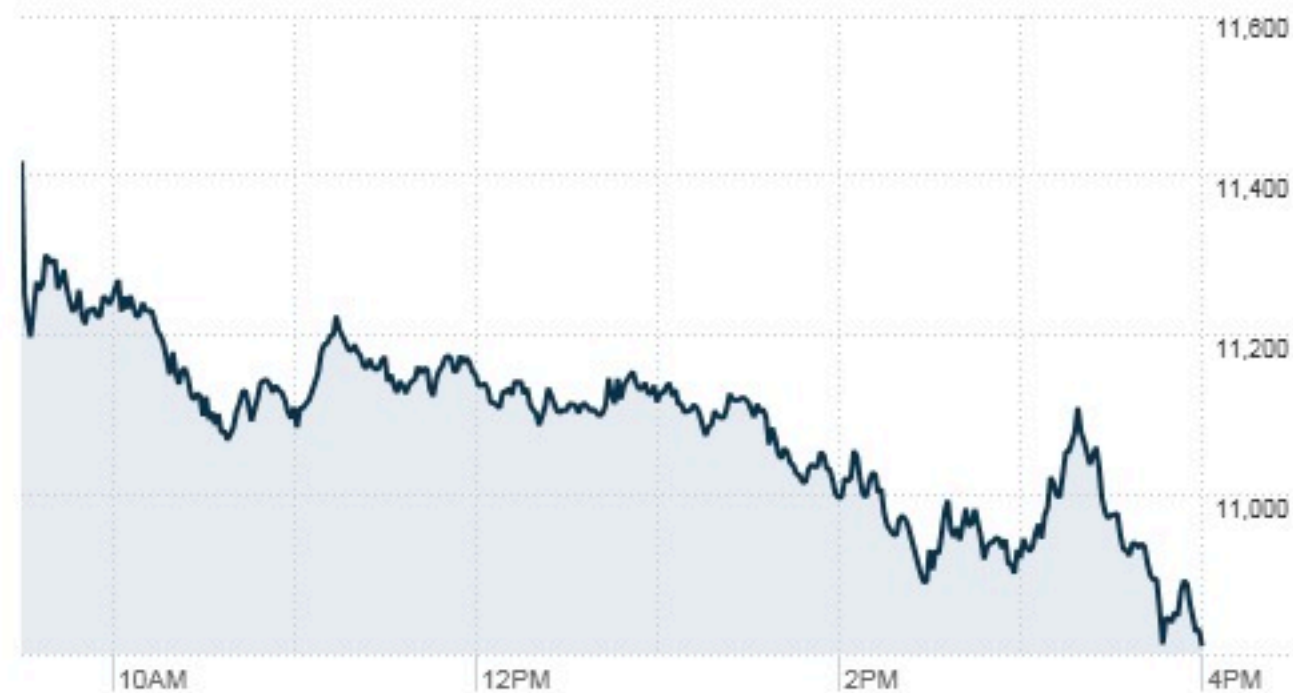
Dow plunges after S&P downgrade

CNNMoney

5,663 comments

By Ken Sweet, contributing writer August 8, 2011: 4:48 PM ET

Dow

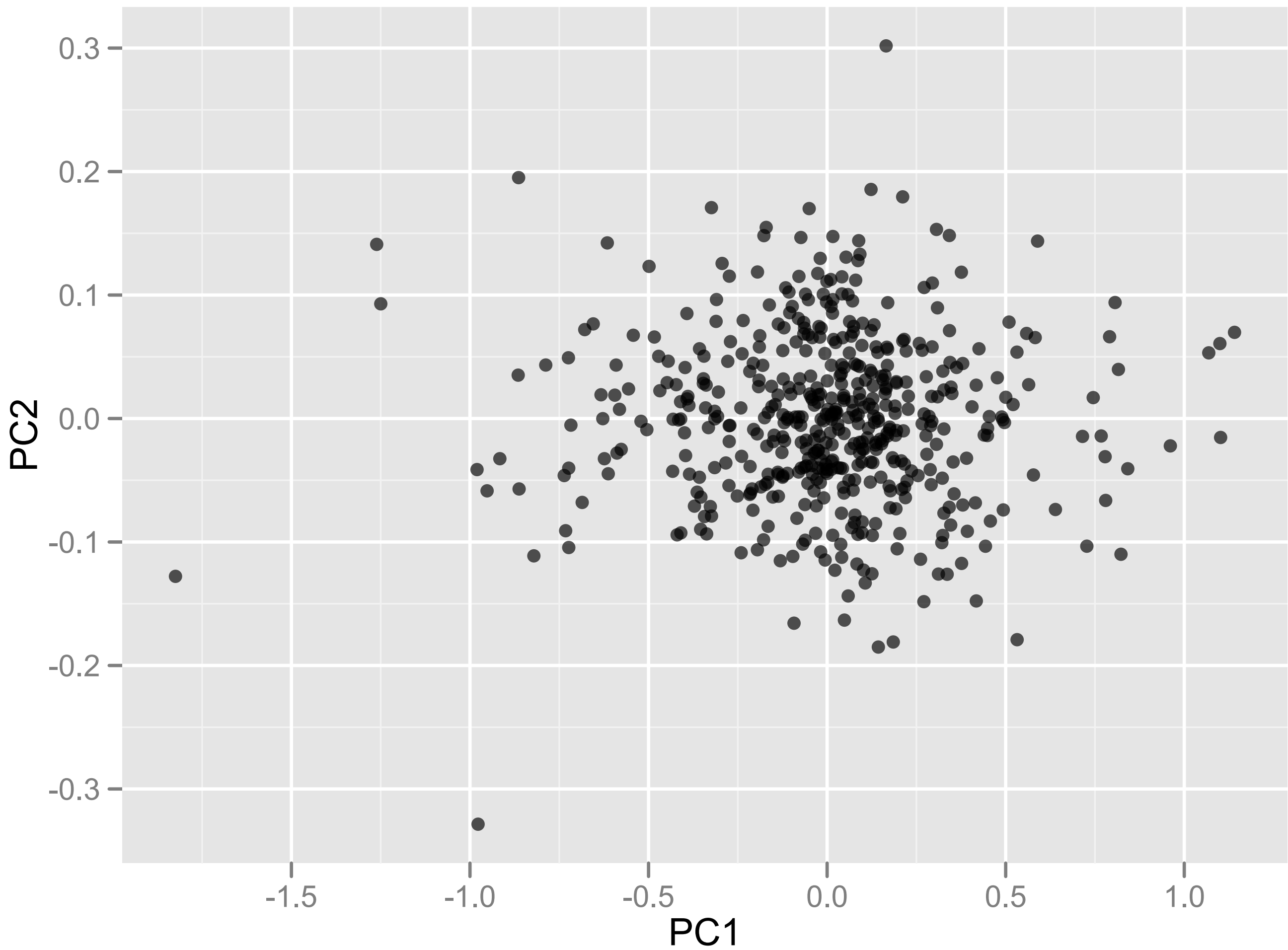


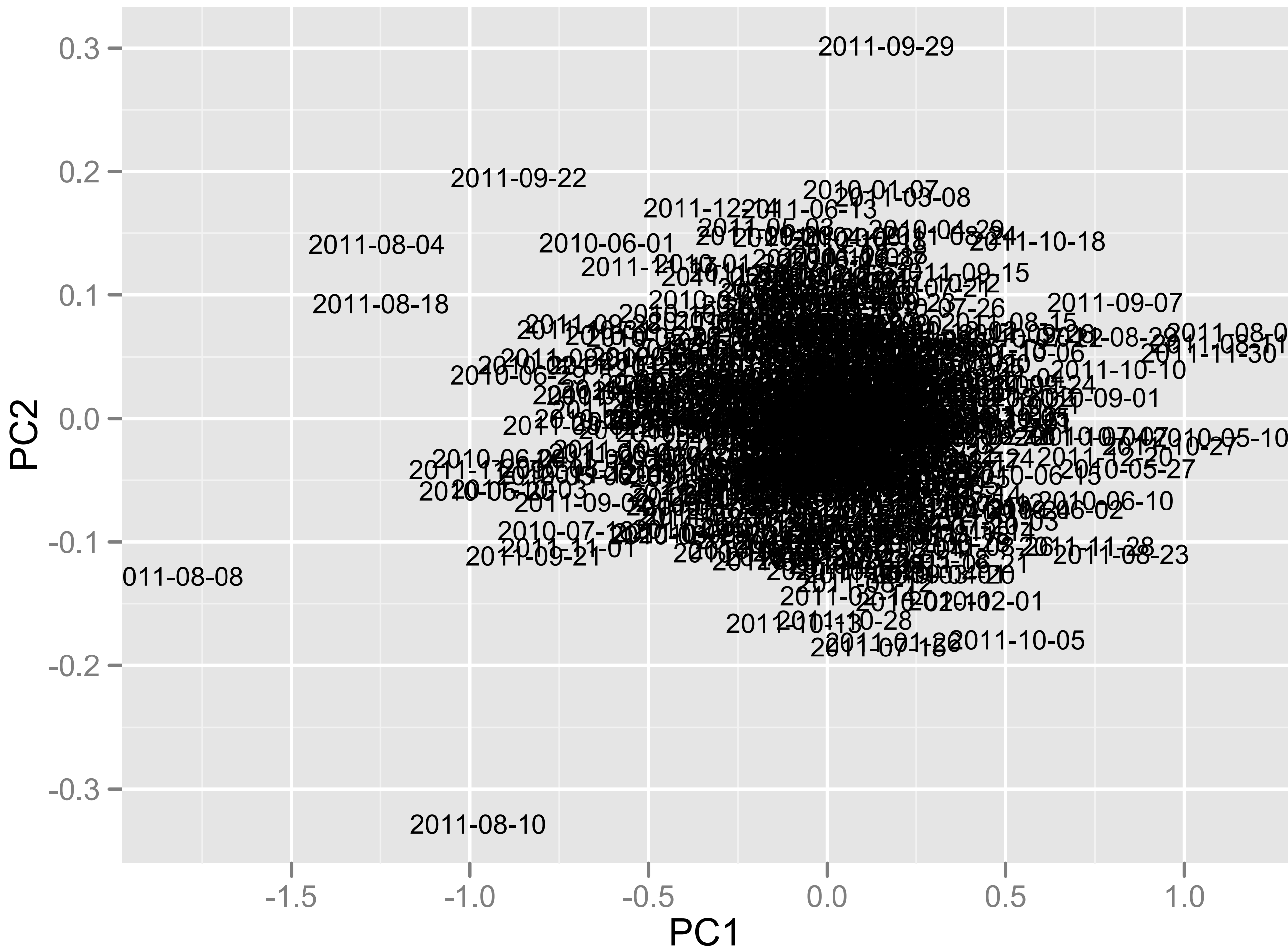
Click the chart for more market data.

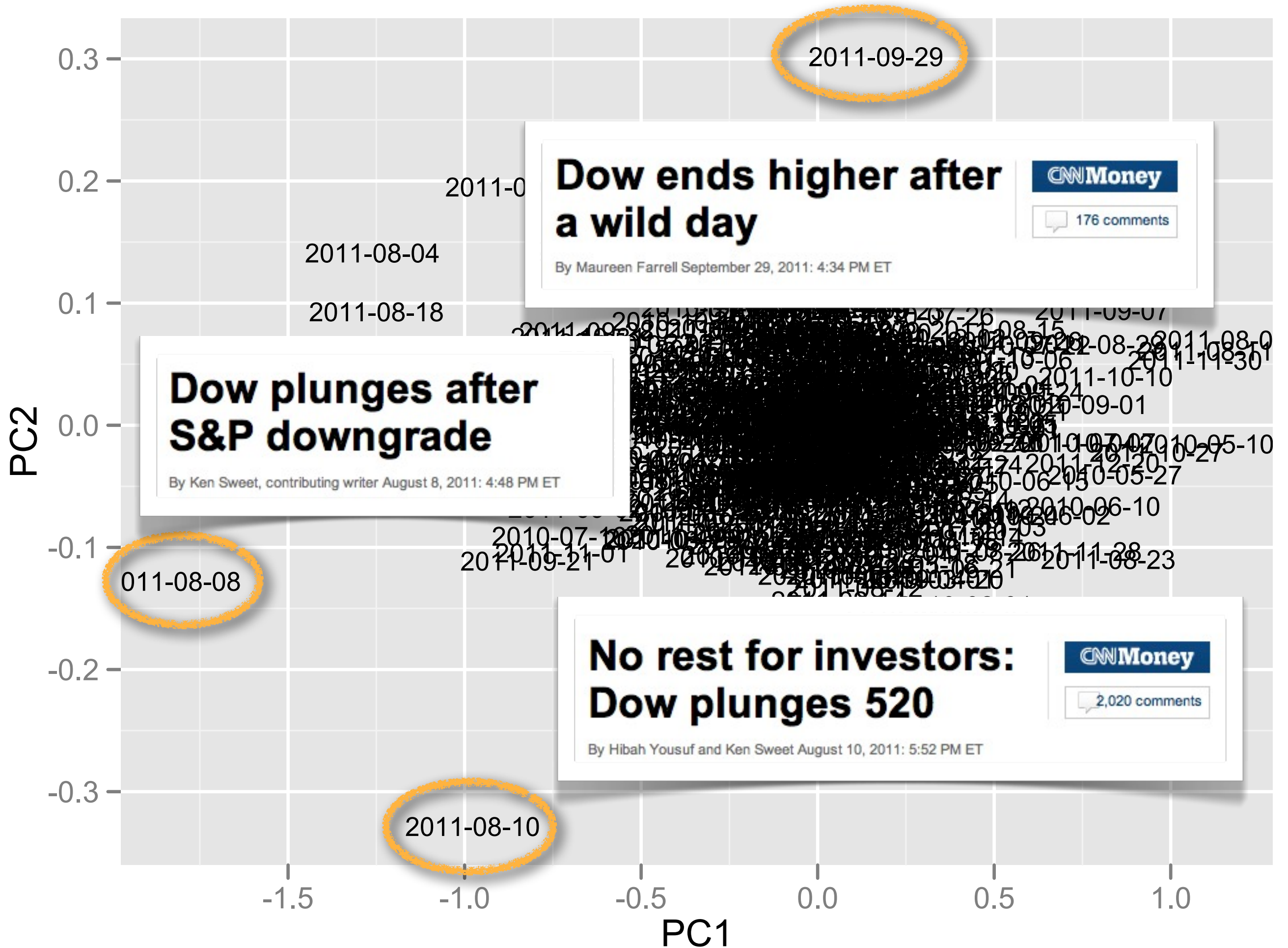
NEW YORK (CNNMoney) -- Wall Street had its worst day since the 2008 financial crisis, as fearful investors reacted to the United States losing its coveted AAA credit rating.

All three major U.S. stock indexes sank between 5% and 7%, pushing the Dow below 11,000 for the first time since last November.

“[August 8, 2011 was the] worst day since the 2008 financial crisis”





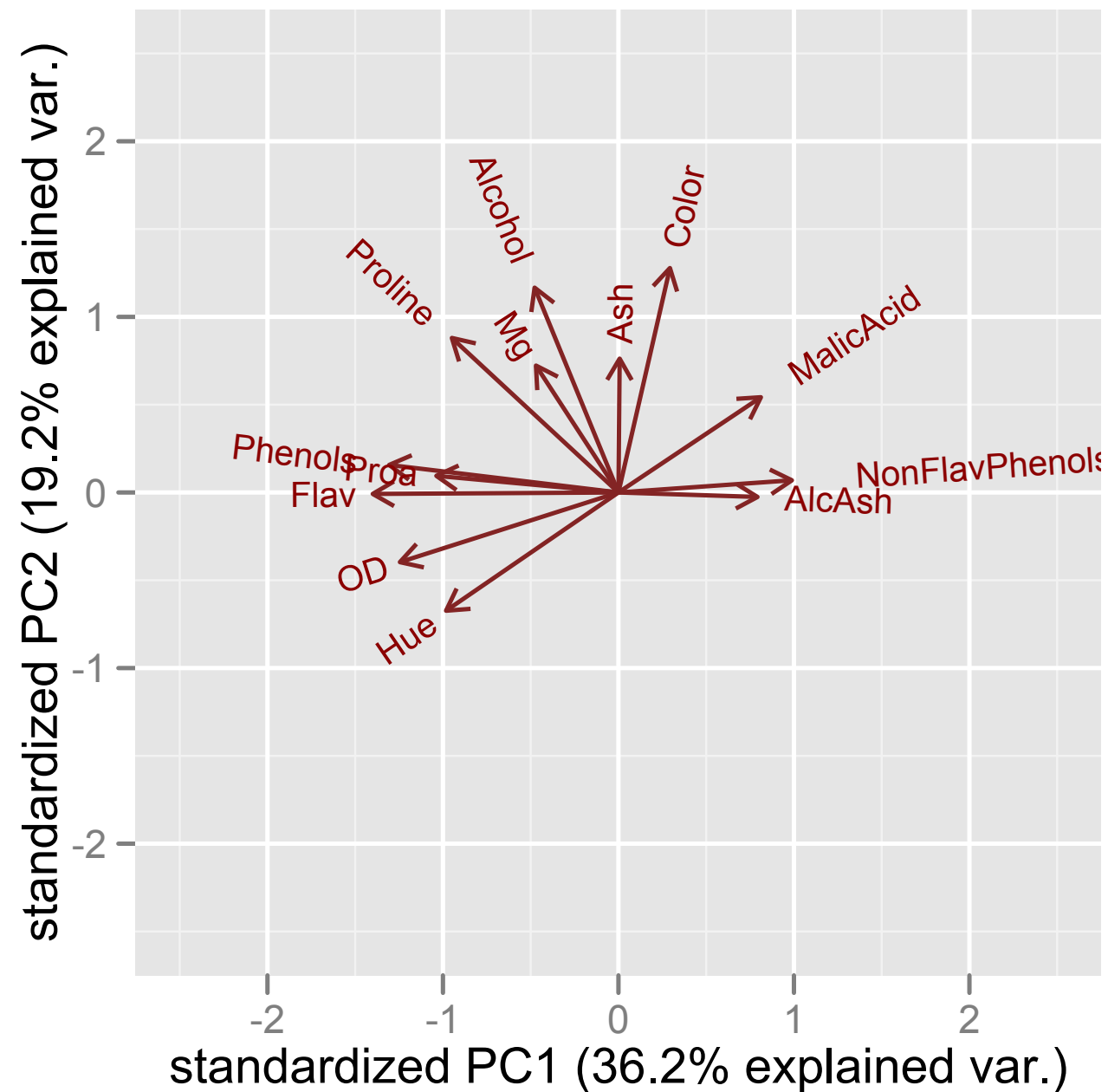


Principal Components Biplot

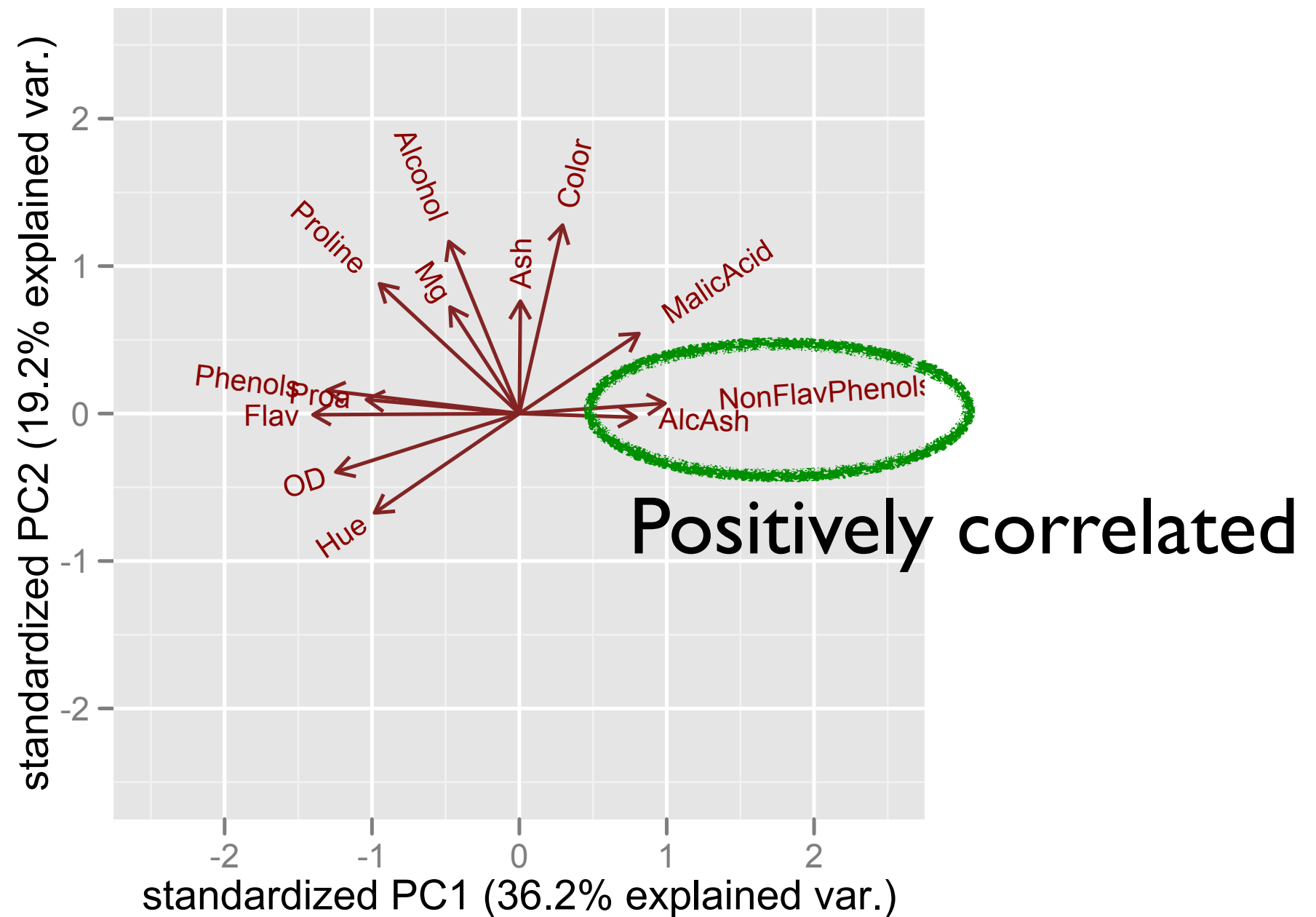
Principal Components Biplot

- Gabriel, K. R. (1971)
- Tool for visualizing variation of the data
- Shows observations and variables ***simultaneously***
- Combines two layers of points
 - observations – principal component scores
 - variables – principal component loadings

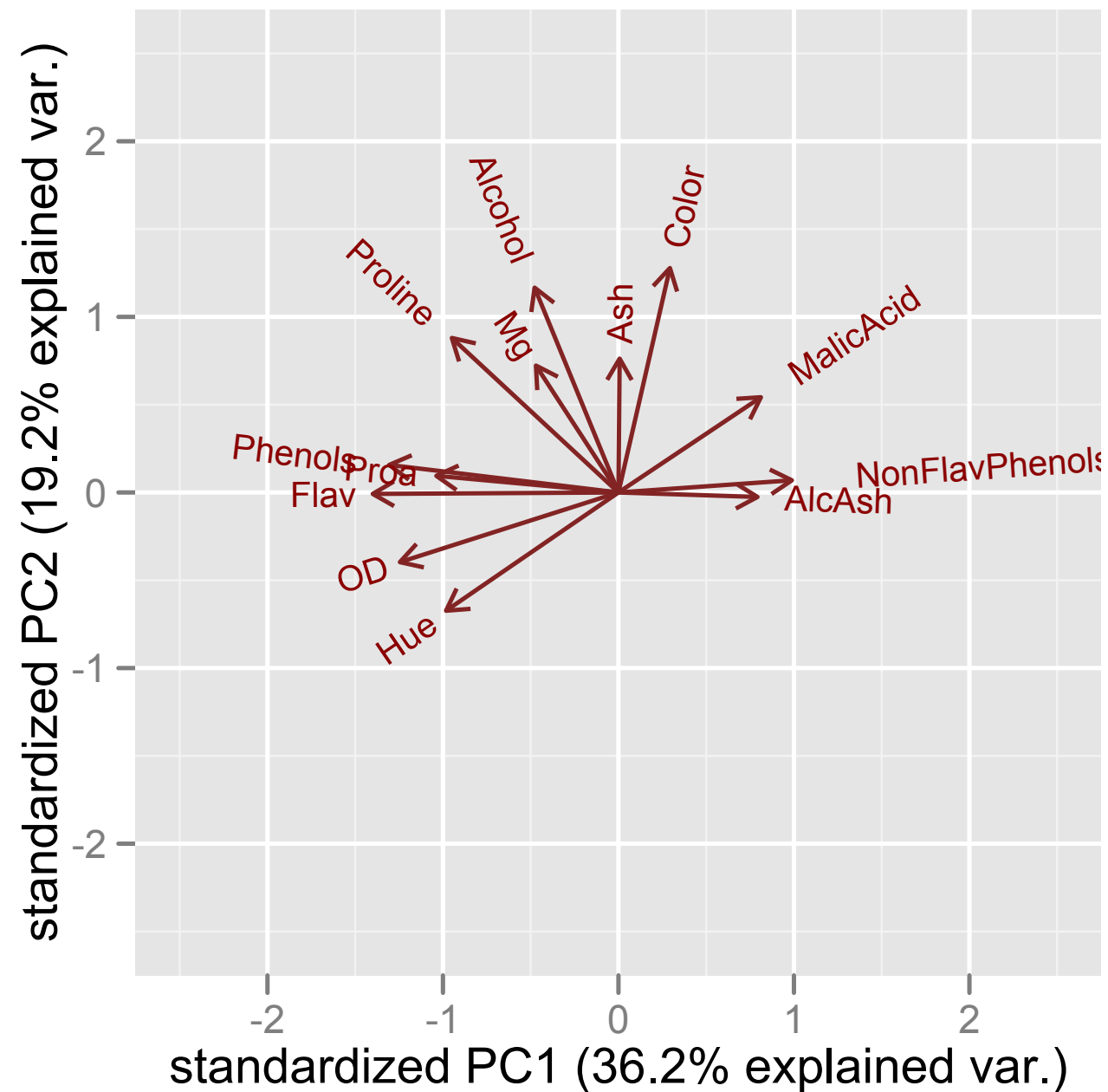
Dot products approximate sample covariance



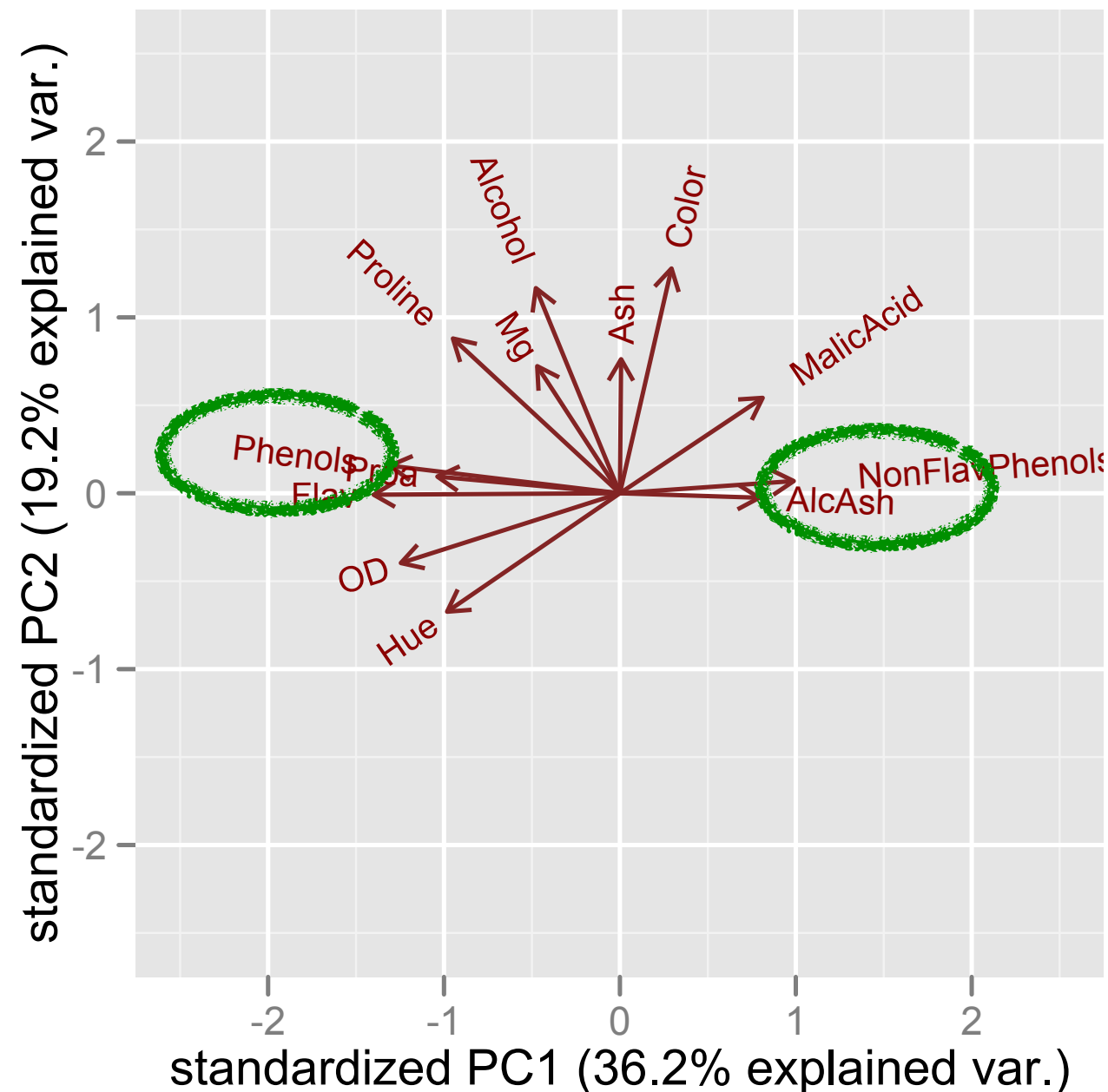
Dot products approximate sample covariance



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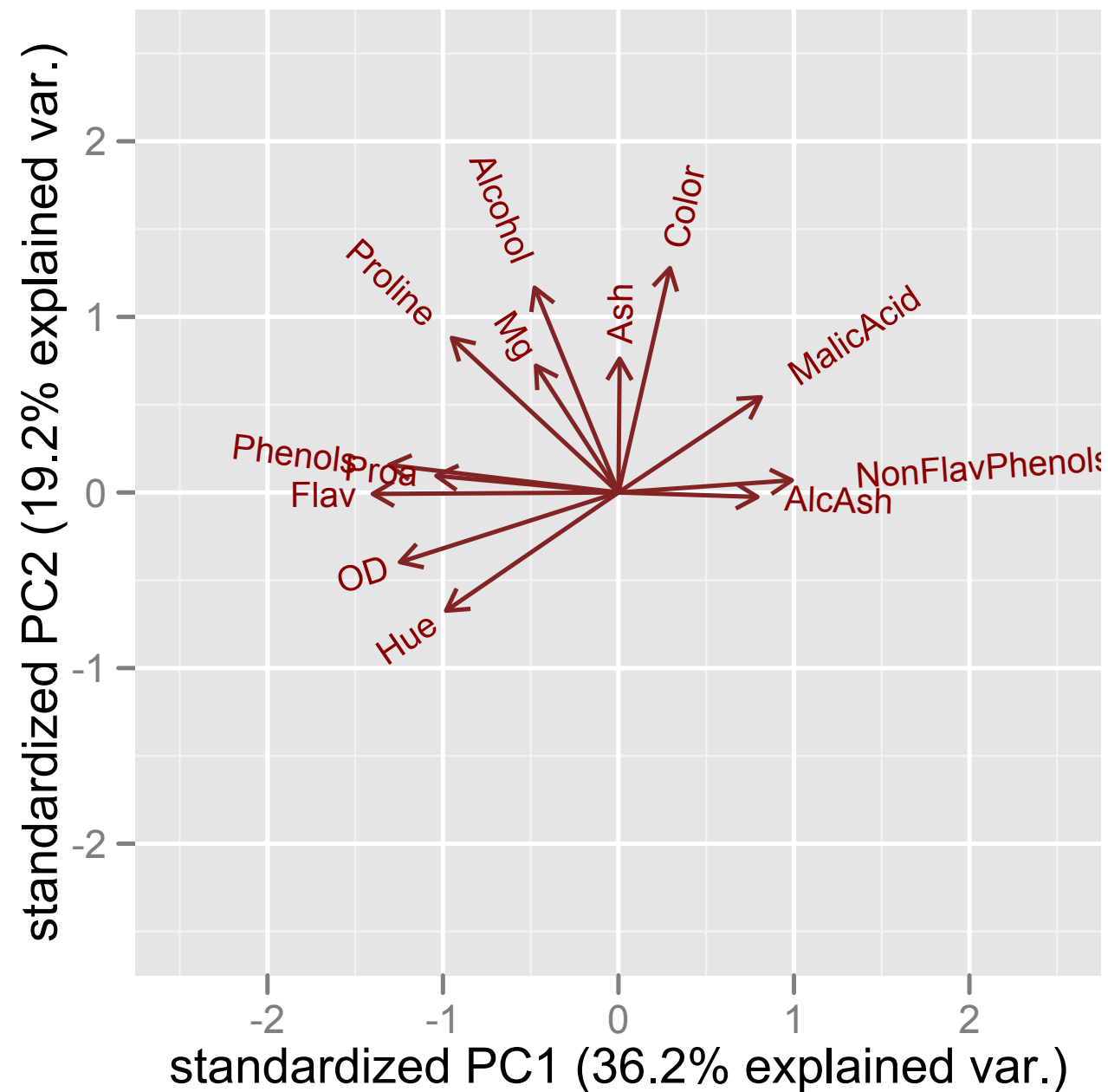


Dot products approximate sample covariance

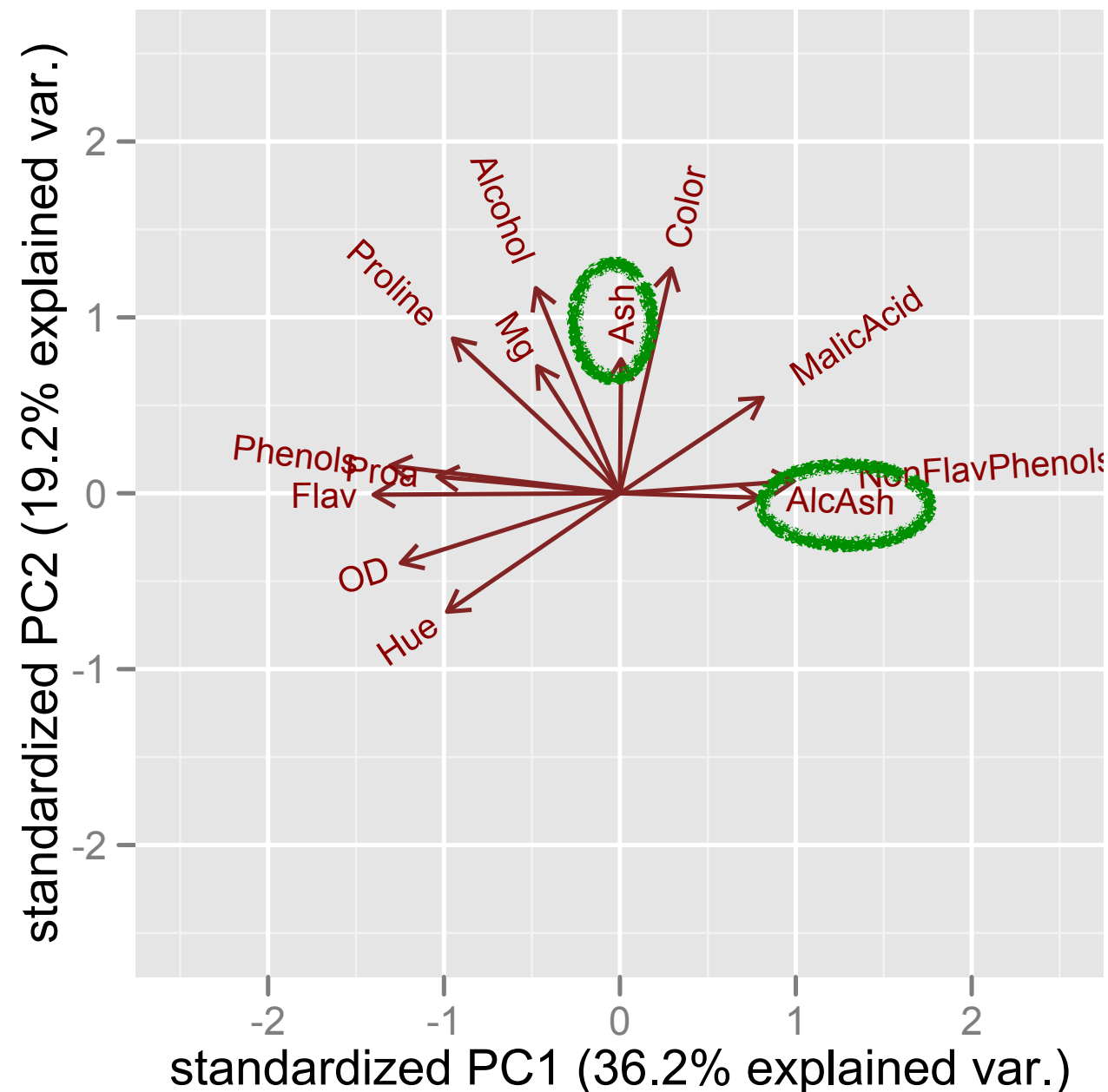


Example: Phenols and AlcAsh are negatively correlated

Dot products approximate sample covariance

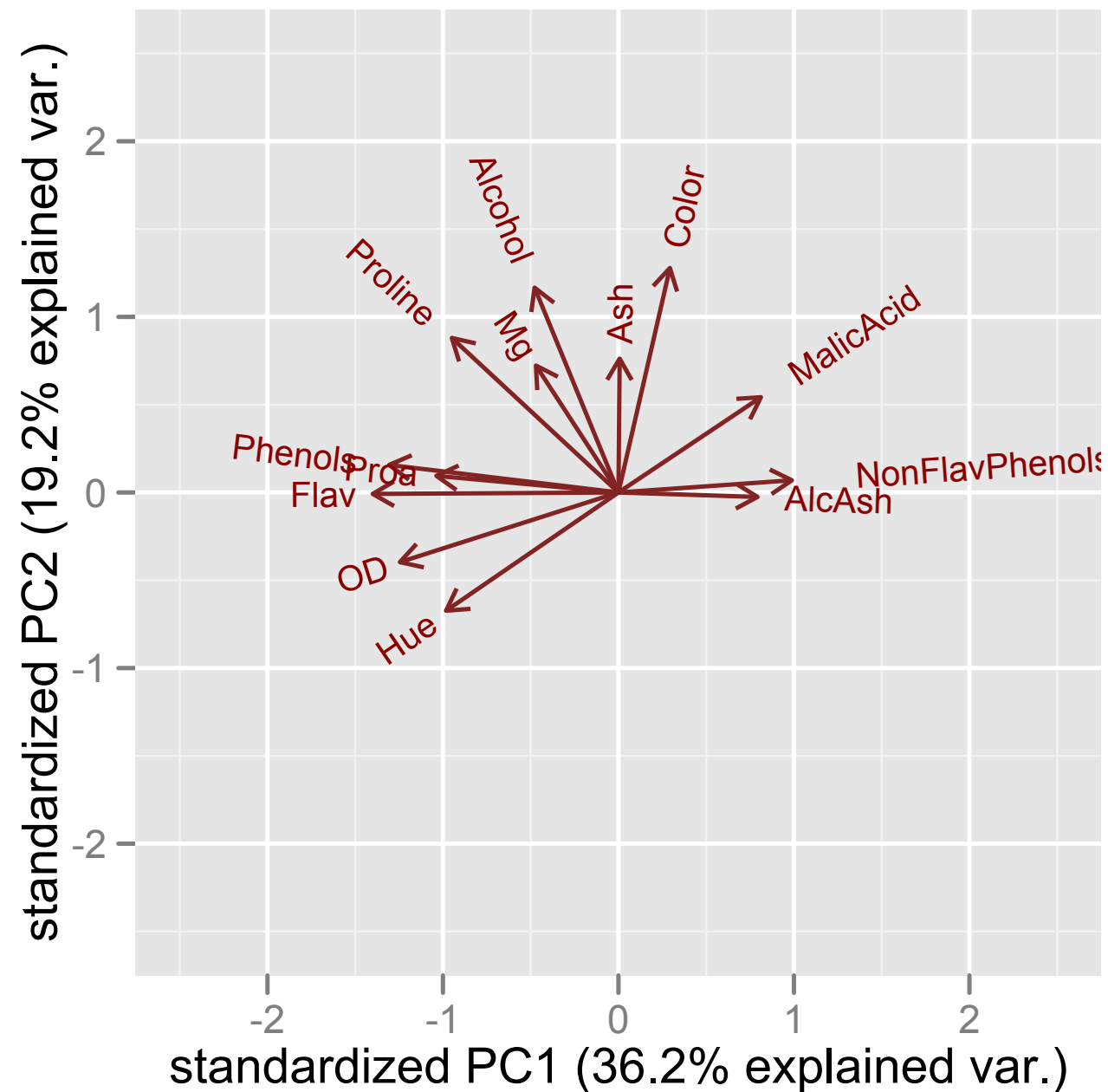


Dot products approximate sample covariance

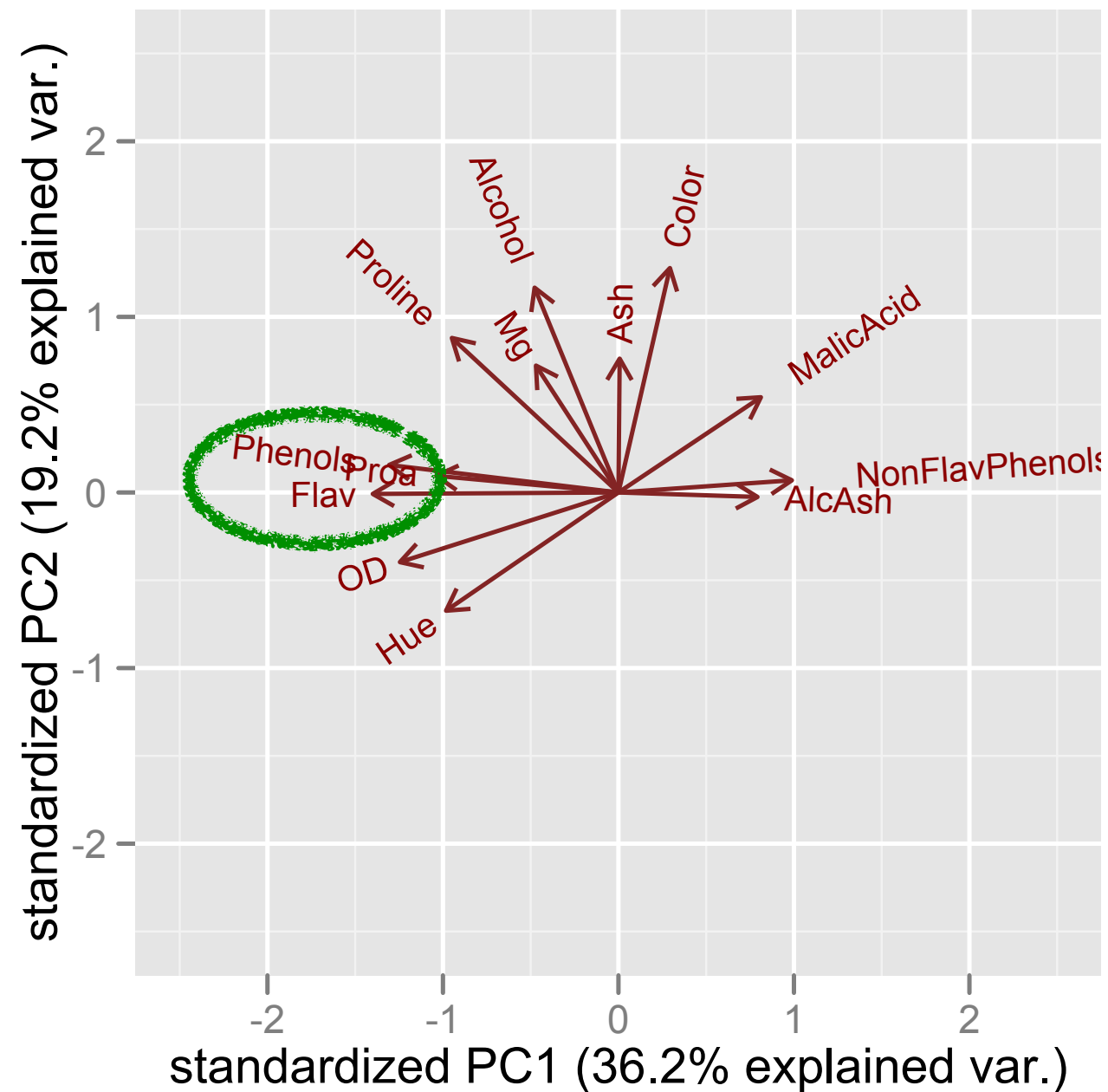


Example: Ash and AlcAsh are nearly uncorrelated

Dot products approximate sample covariance

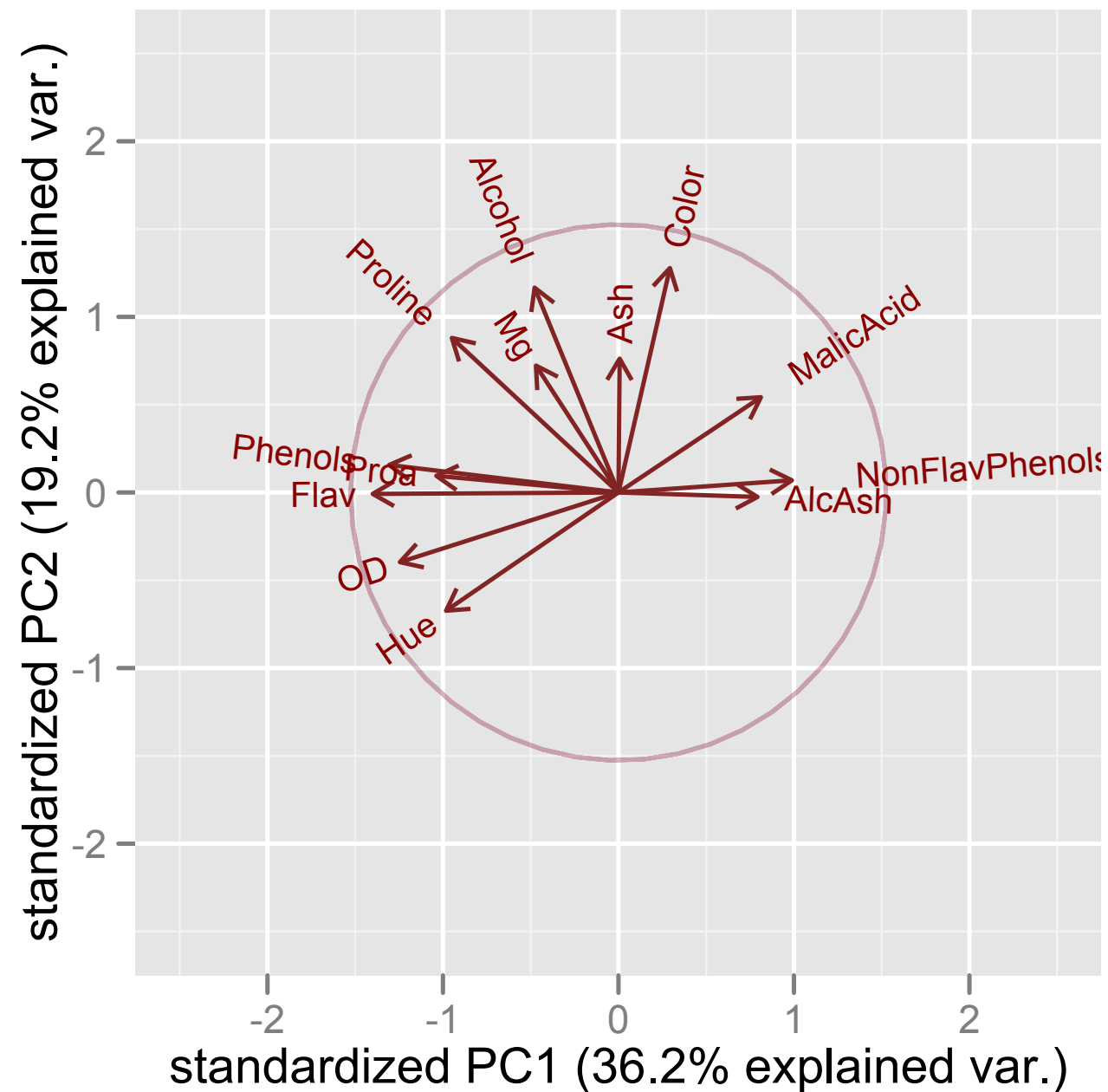


Dot products approximate sample covariance

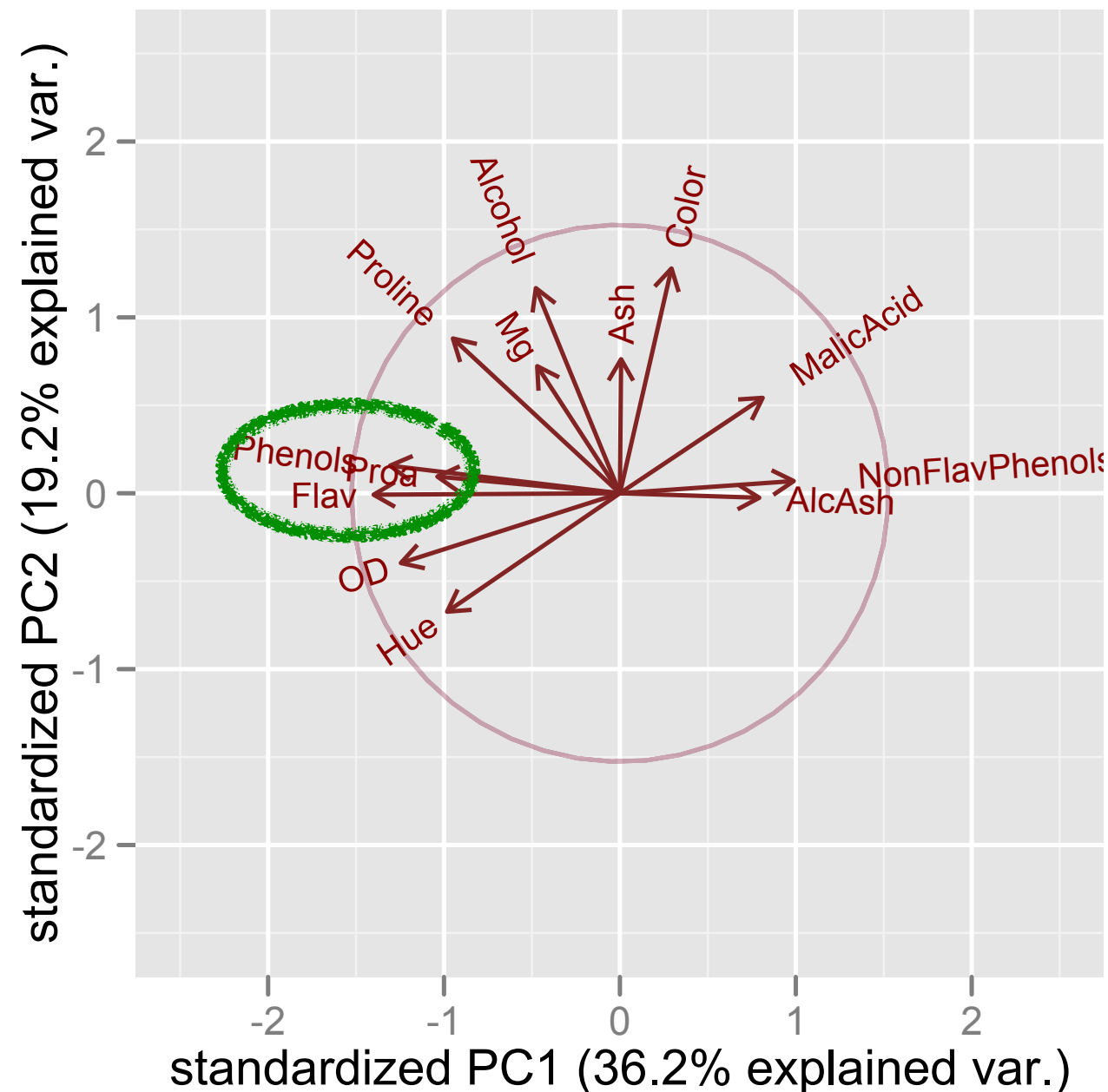


Example: Phenols, Proa, and Flav are positively correlated.

Circle: theoretical maximum

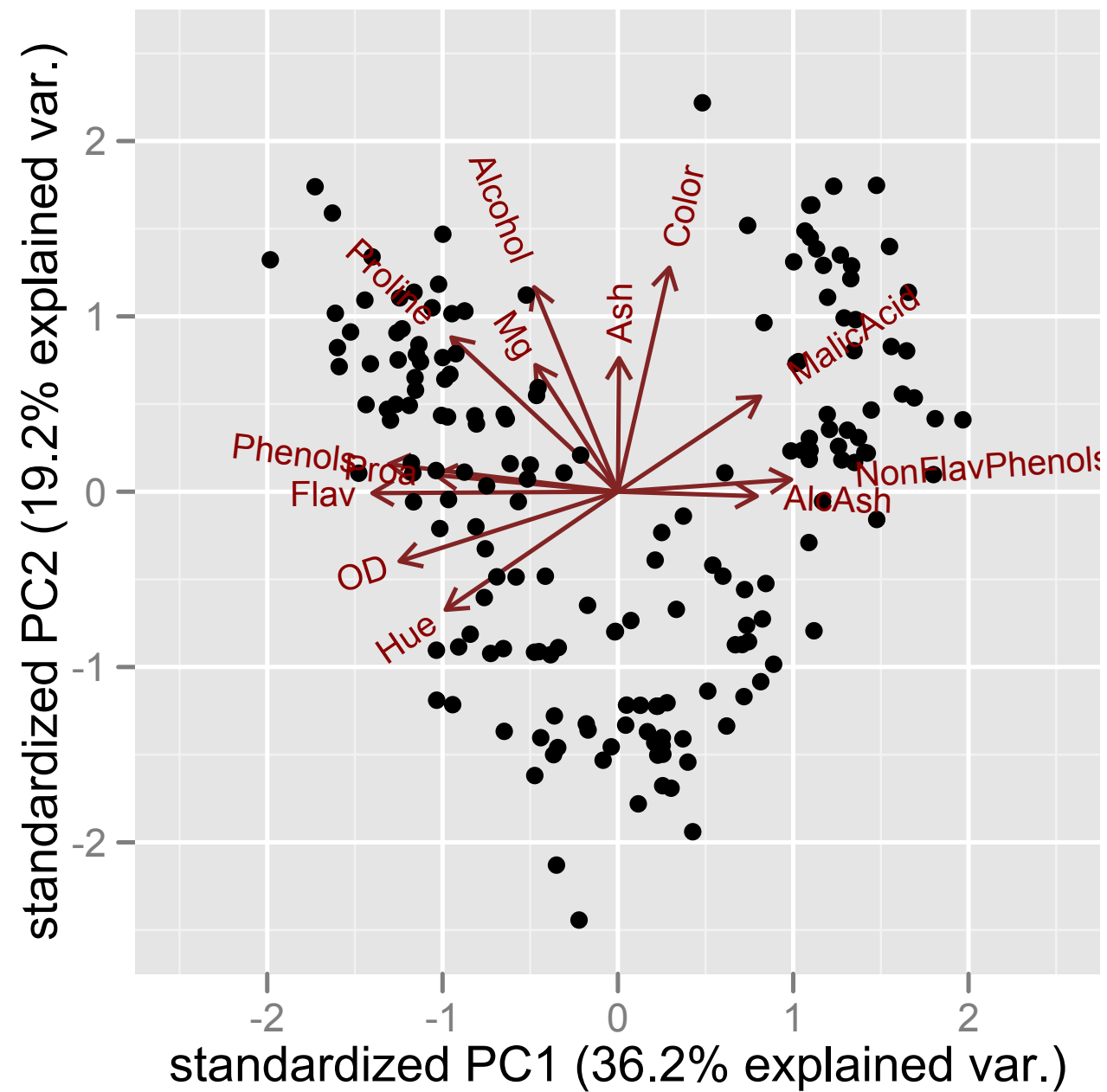


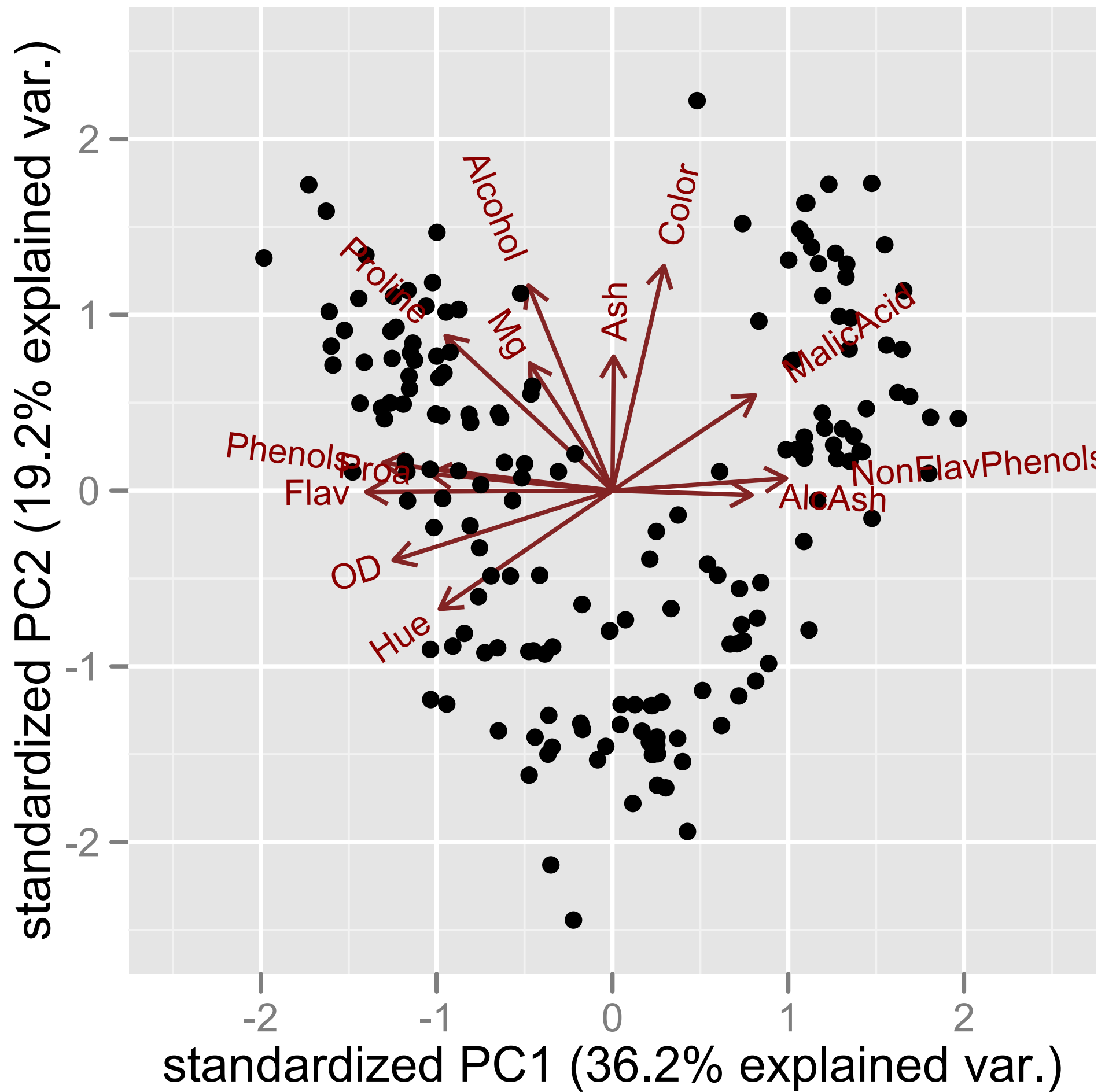
Circle: theoretical maximum

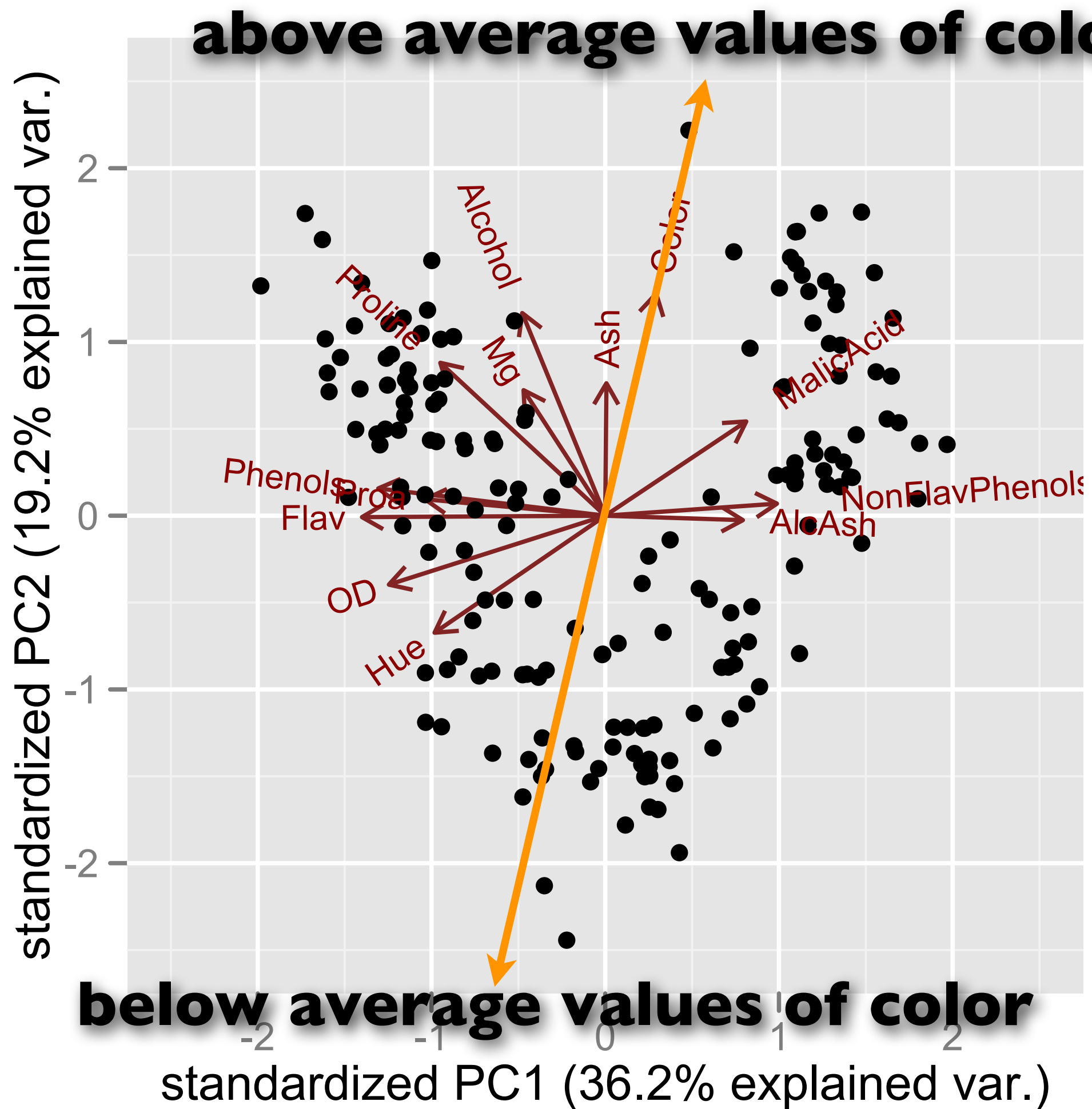


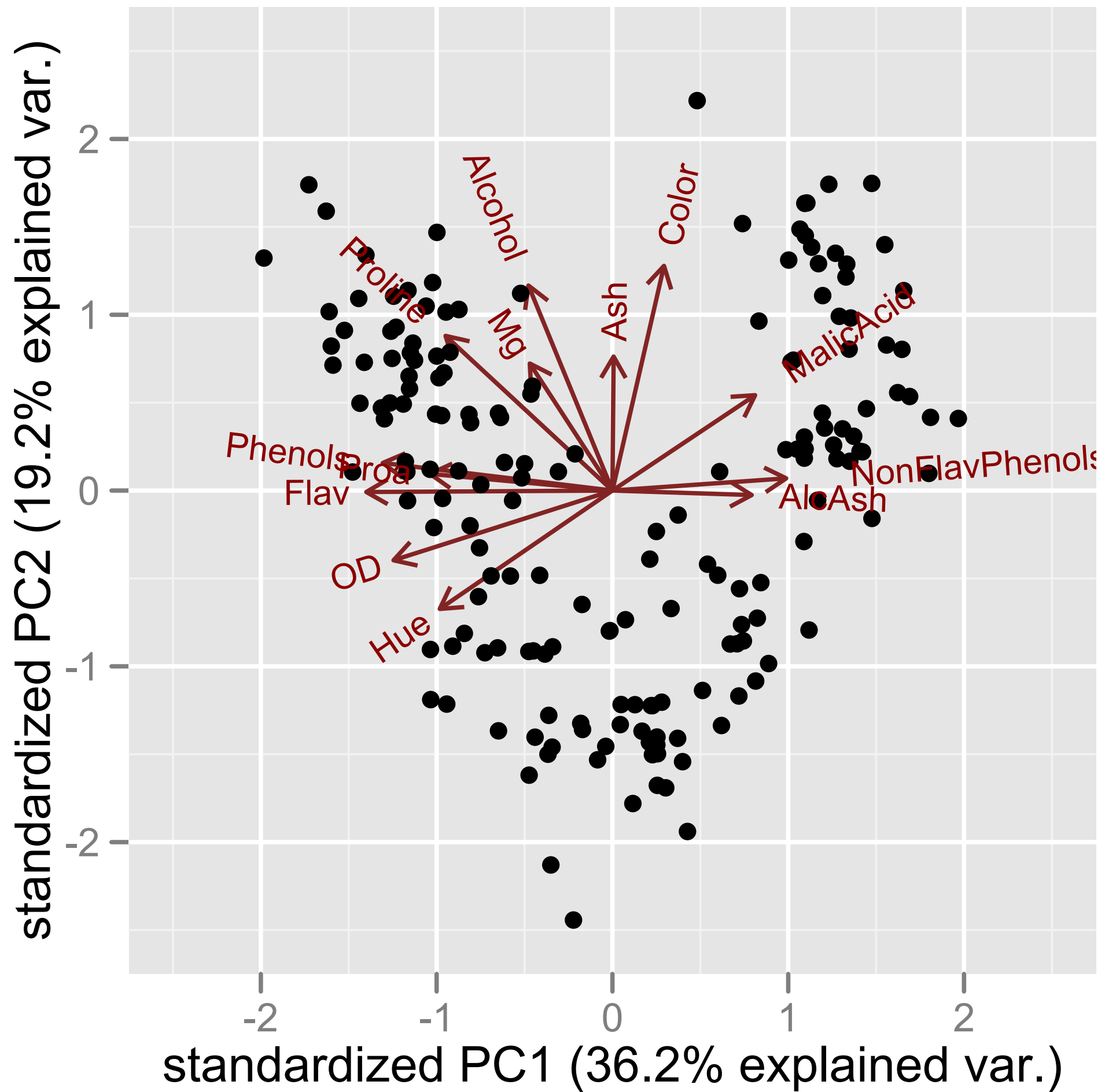
Phenols, Proa, and Flav load mostly on PC1

Arrows are axes

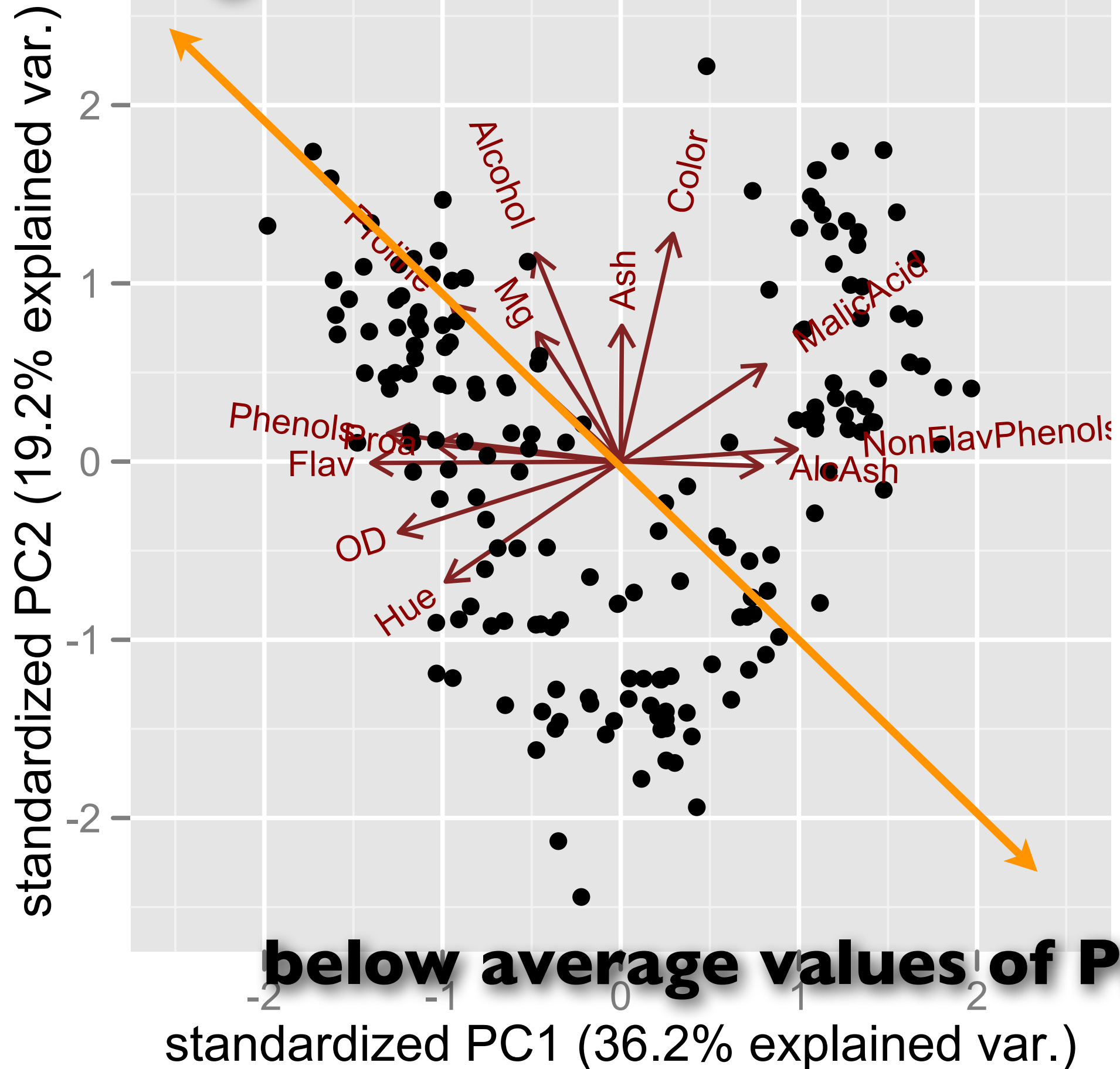


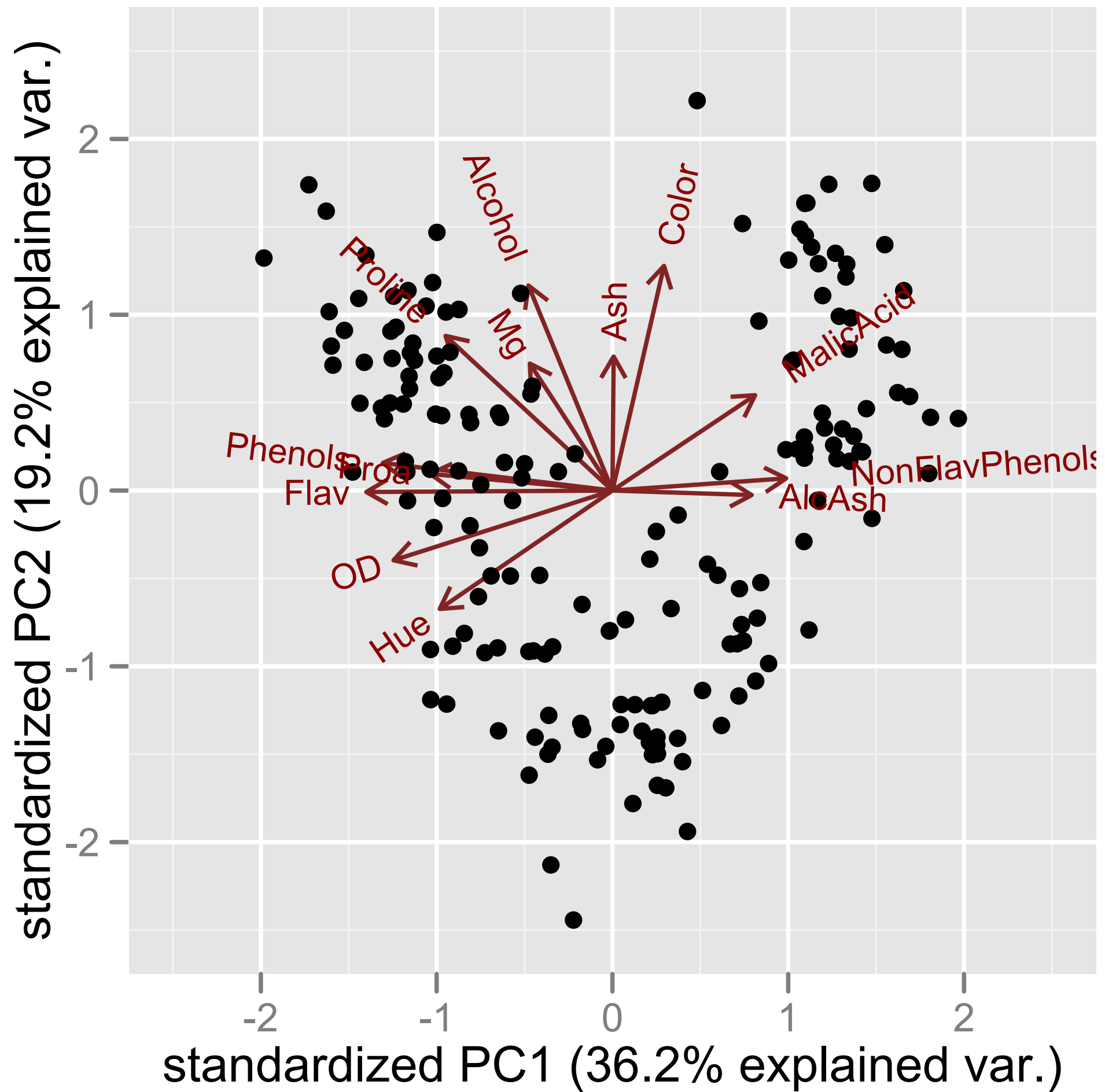


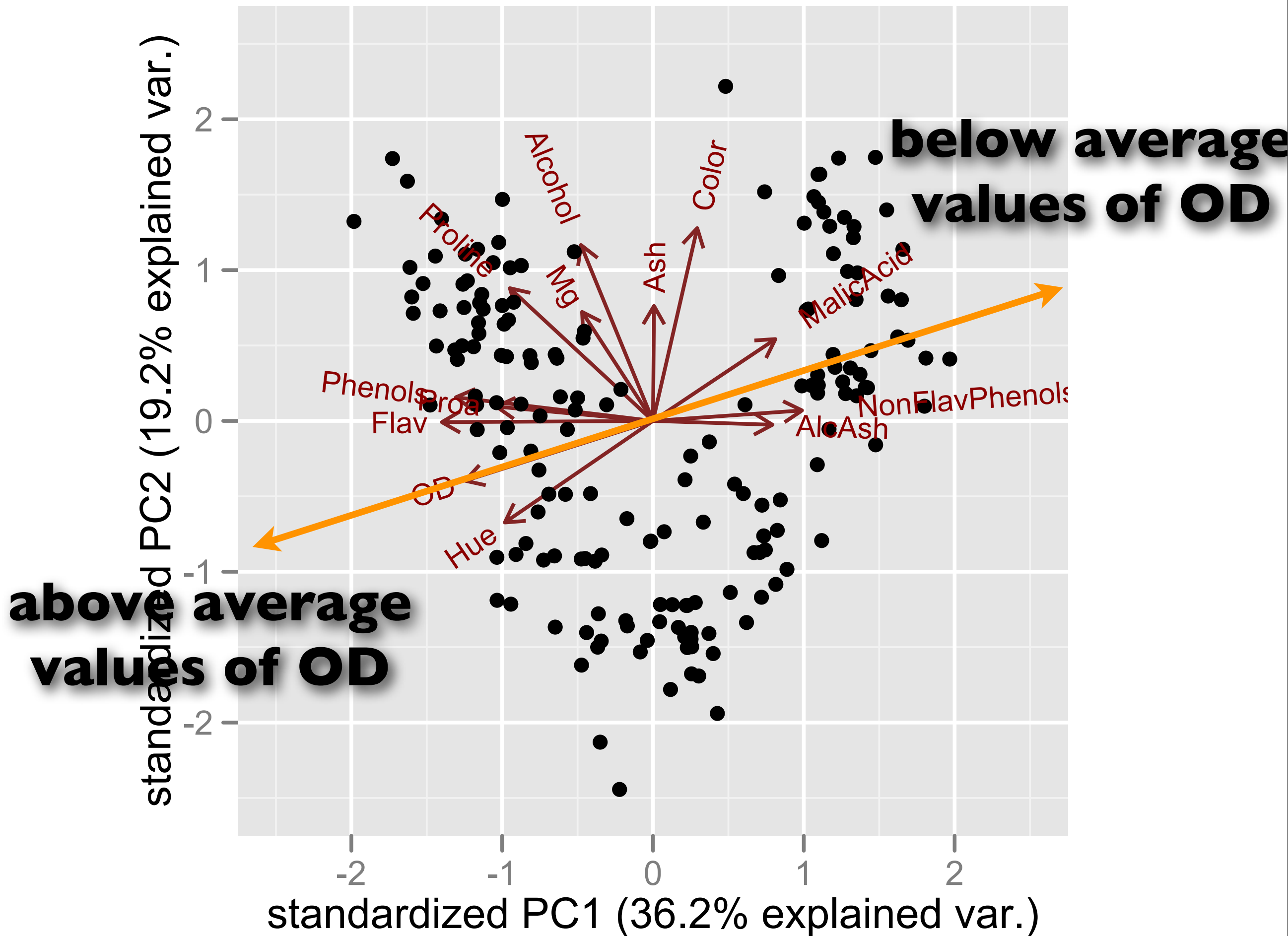


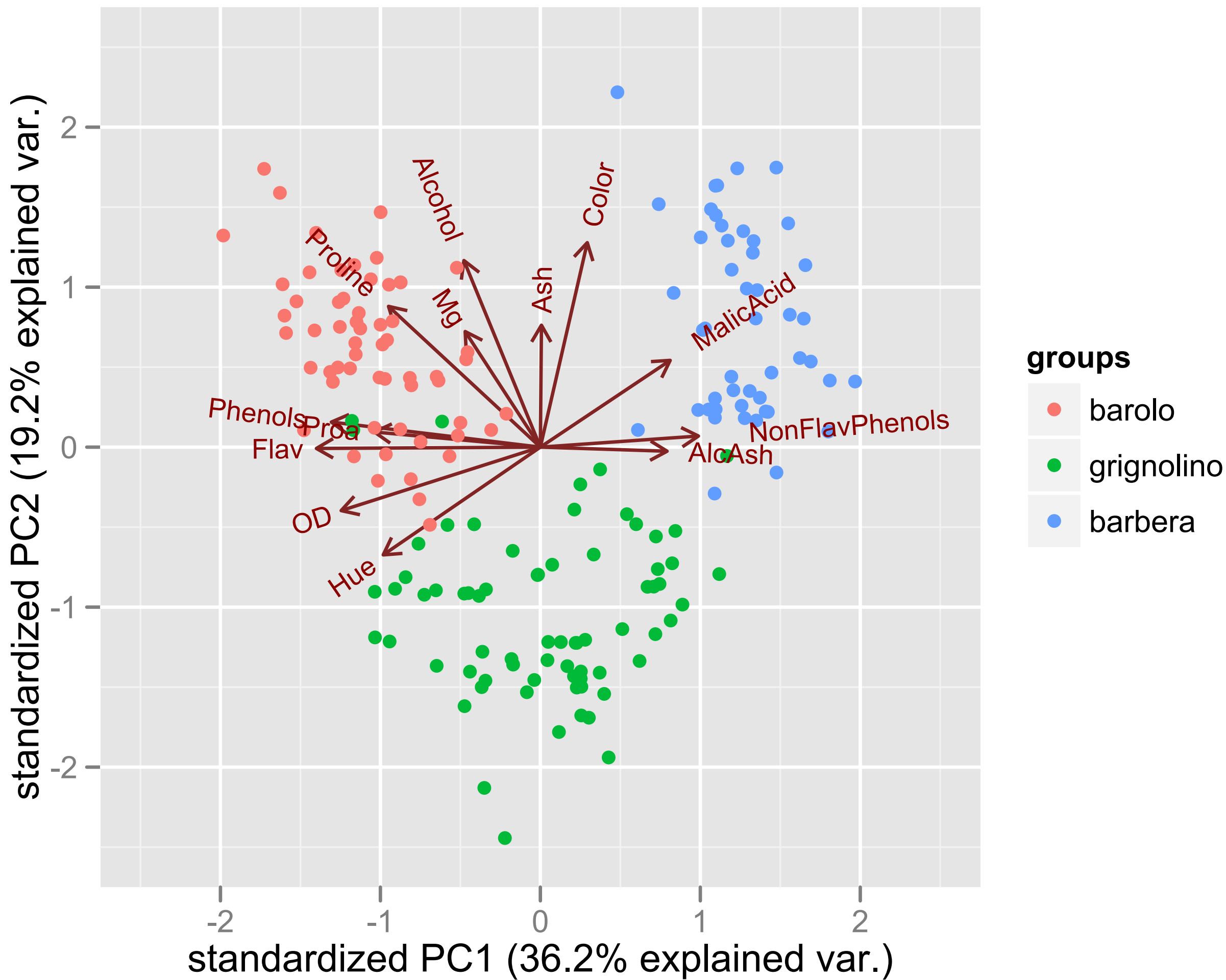


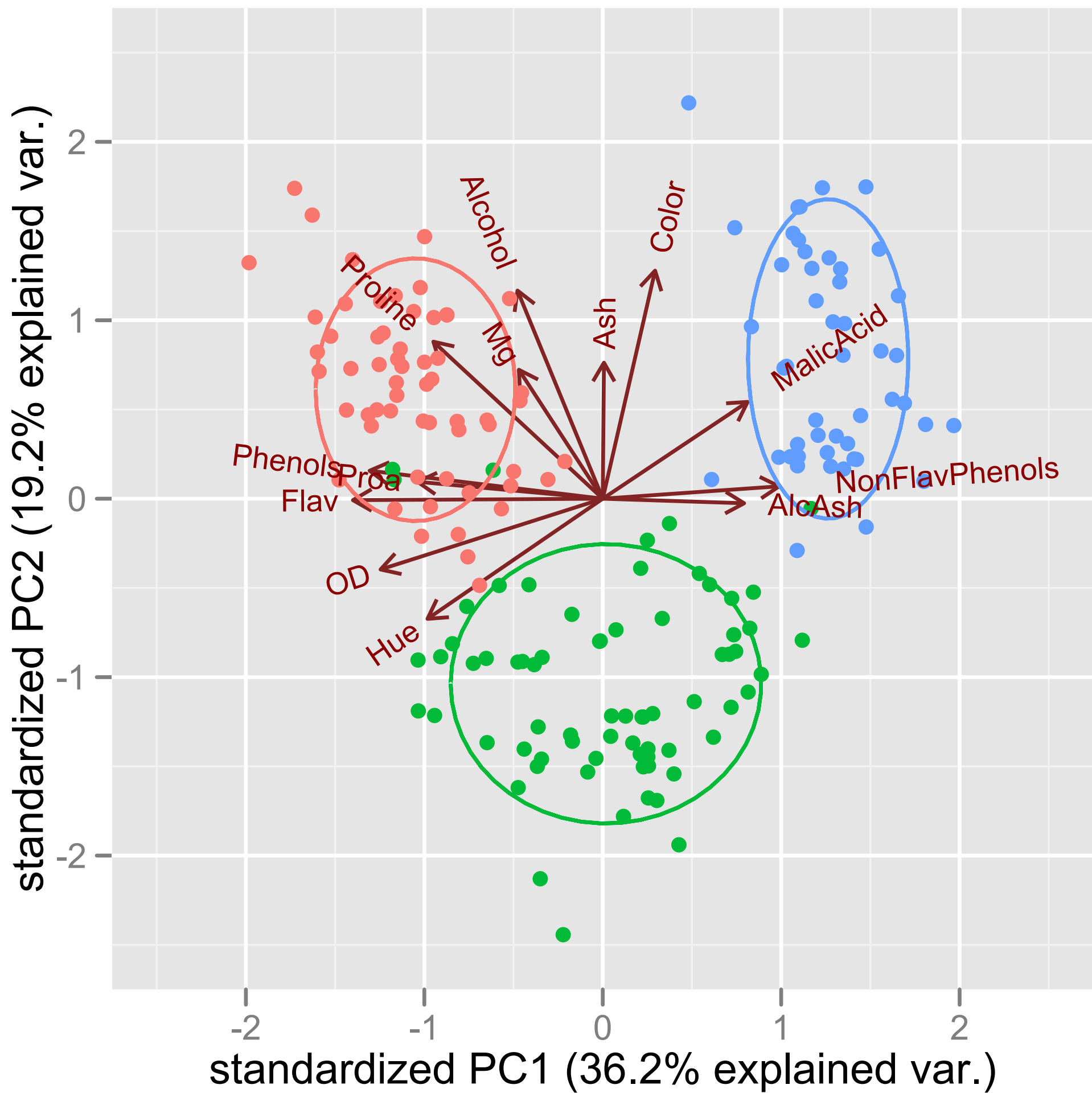
above average values of Proline











Implementations

- R built-in: `biplot()`
- <http://github.com/vqv/ggbiplot>